$t\bar{t}$ and $hh \to b\bar{b}\,\ell\nu\,q\bar{q}'$: samples, the ten angles, and $f_{t\bar{t}}$ by OSDE

Tae Min Hong

PITT PACC and Department of Physics and Astronomy, University of Pittsburgh

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Outline

1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Spherical-harmonic basis
9. Is the OSDE approach helping? (ML)
10. Why the separation washes out at gen level
11. Adding momentum to y: the resonance masses
12. Discriminator t (paper style)
13. Acceptance
14. Study A′: hh with on-shell W bosons ($m_H = 170$ GeV)
15. Study B: $t\bar{t}$ vs $H \to t\bar{t}$
16. Pileup
17. Level-1 trigger: a lookup-table version of angles + BW
18. Summary

Goal

- Generate $pp \to t\bar{t} \to b\bar{b}\,W^+W^- \to b\bar{b}\,\ell\nu\,q\bar{q}'$ at parton, particle (gen) and detector (reco) level
- Validate the samples and the ten angles $\mathbf{x} = \{Y, \Theta, \boldsymbol{\theta}, \tilde{\boldsymbol{\theta}}\}$ used for the $t\bar{t}$ density $f_{t\bar{t}}(\mathbf{x})$
- Everything is scripted: `generate_samples.py`, `make_plots.py`, `make_slides.py`

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Sample generation

- MadGraph5_aMC@NLO 3.6.7 (LO, `sm`, 4-flavour proton) $\rightarrow$ Pythia8 $\rightarrow$ Delphes 3.5.1, ATLAS card, **no pileup**
- Decay chains (spin correlations kept), no generator-level cuts: $pp \rightarrow t\bar{t}$, $t \rightarrow bW^+(\rightarrow \ell^+\nu)$, $\bar{t} \rightarrow \bar{b}W^-(\rightarrow q\bar{q}')$, plus the charge conjugate
- $\sqrt{s} = 13$ TeV, $\ell = e, \mu$, 10000 events, seed 12345
- $\sigma = 136.5$ pb $\approx \sigma_{t\bar{t}}^{\mathrm{LO}} \times 2\,\mathcal{B}(W \rightarrow \ell\nu)\,\mathcal{B}(W \rightarrow q\bar{q}')$ (with $\ell = e, \mu$)

Event levels

	parton	parton + ℓ trig.	gen (particle)	reco
source	MG5 LHE	MG5 LHE	Pythia8	Delphes ATLAS
selection	none	ℓ cuts only	$1\ell, \geq 4j, \geq 1b, \chi^2$	$1\ell, \geq 4j, \geq 1b, \chi^2$
events kept	100%	71.7%	41.3%	24.4%
E_T^{miss}	p_T^{ν}	p_T^{ν}	$\sum$ all ν	$-\sum$ calo towers

Parton = MG5 partons with *no* cuts; parton + ℓ trigger adds only the lepton cuts ($p_T > 25$ GeV, $|\eta| < 2.47$ (e), 2.5 (μ)). Reco and gen E_T^{miss} agree for muon and electron events (reco−gen along the lepton: -5.9 vs -5.2 GeV), so the calorimeter MET does not lose muons.

Object definitions (gen and reco)

	particle level (gen)	reco (Delphes ATLAS)
jets	GenJet anti-k_t 0.4	Jet anti-k_t 0.4
	$p_T > 25$ GeV, $\lvert\eta\rvert < 2.5$	
b-jets	$\Delta R(j,b) < 0.4$, $p_T^b > 5$ GeV	BTag bit 0
leptons	status-1 e/μ from $W/Z/h$ (or such a τ)	Electron, Muon
	$p_T > 25$ GeV, $\lvert\eta\rvert < 2.47$ (e), 2.5 (μ)	
overlap	drop jets with $\Delta R(j,\ell) < 0.4$	UniqueObjectFinder

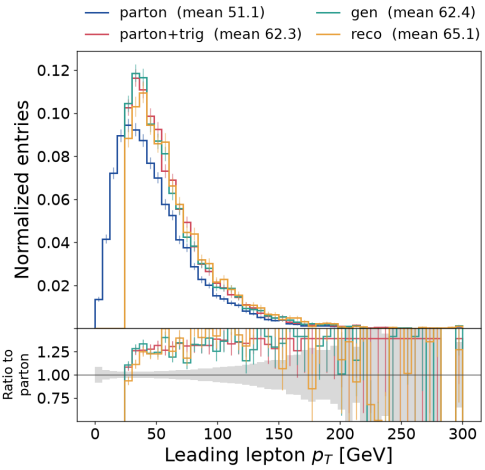
Preselection: exactly 1 lepton, ≥ 4 jets, ≥ 1 b-jet, then the χ^2 $t\bar{t}$ reconstruction.

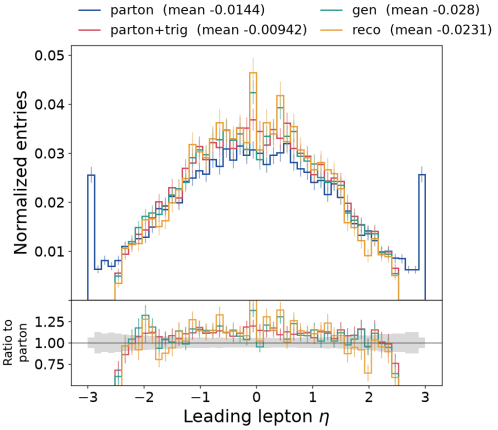
Cutflow

selection	gen		reco	
All events	10000	100.0%	10000	100.0%
exactly 1 lepton	7071	70.7%	4907	49.1%
≥ 4 jets	4196	42.0%	2830	28.3%
≥ 1 b-jet	4161	41.6%	2454	24.5%
≥ 2 b-jets	3305	33.0%	1221	12.2%
χ^2 reconstruction (from ≥ 1 b)	4134	41.3%	2440	24.4%

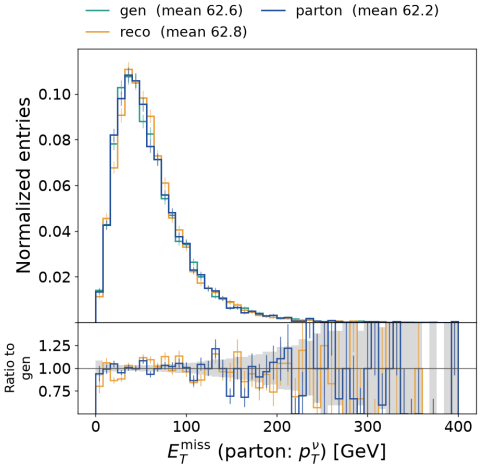
Fractions relative to all generated events.

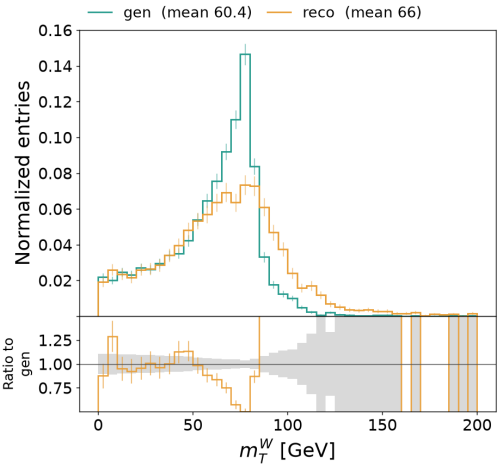
Leptons and E_T^{miss} (1/2)



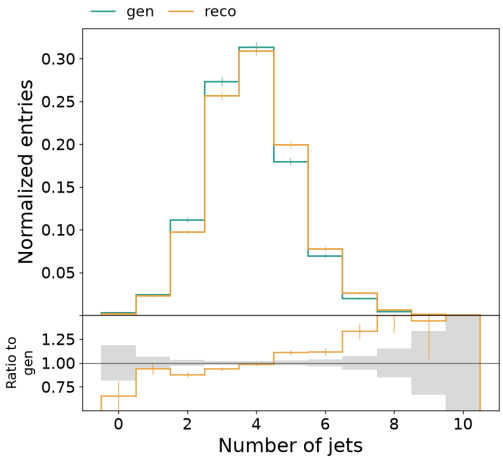


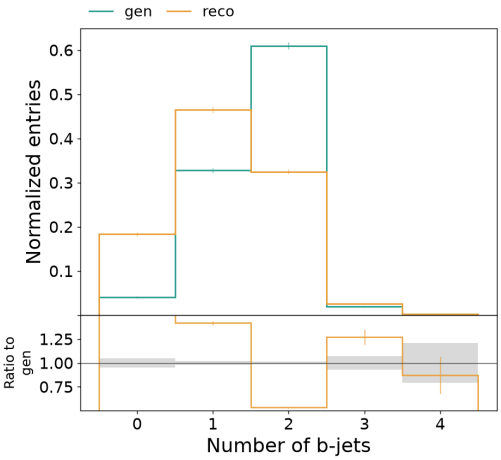
Leptons and E_T^{miss} (2/2)



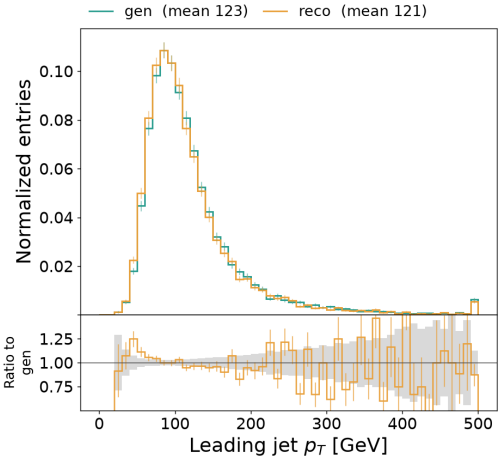


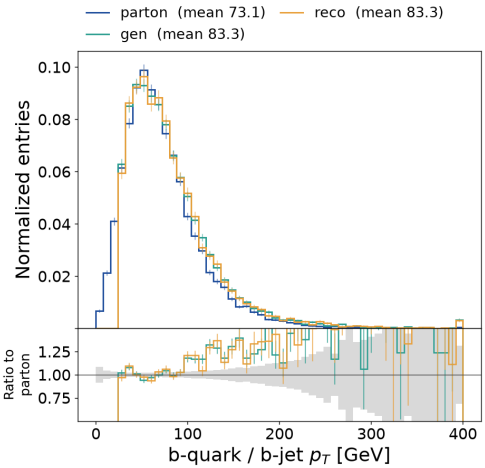
Jets (1/2)


gen
reco
Normalized entries
0.30
0.25
0.20
0.15
0.10
0.05
Ratio to gen
1.25
1.00
0.75
0
2
4
6
8
10
Number of jets


gen
reco
Normalized entries
0.6
0.5
0.4
0.3
0.2
0.1
Ratio to gen
1.25
1.00
0.75
0
1
2
3
4
Number of b-jets

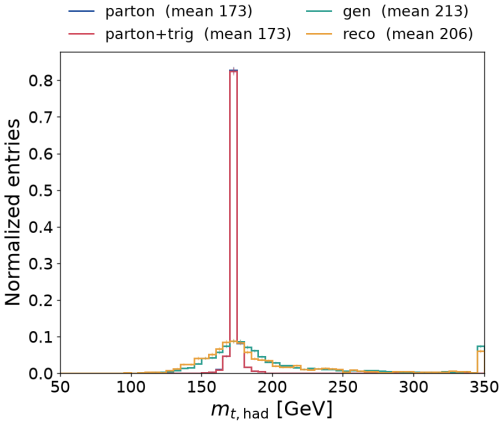
Jets (2/2)

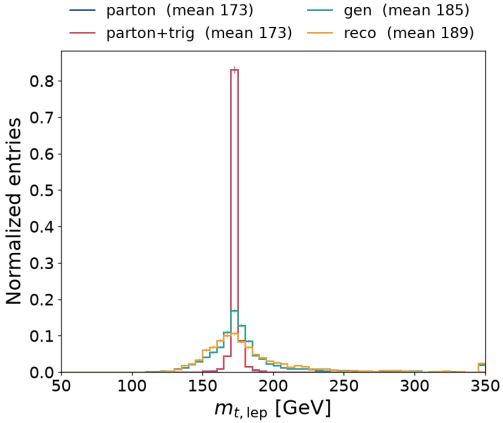

gen (mean 123)
reco (mean 121)
Normalized entries
0.10
0.08
0.06
0.04
0.02
Ratio to gen
1.25
1.00
0.75
0
100
200
300
400
500
Leading jet p_T [GeV]


parton (mean 73.1)
reco (mean 83.3)
gen (mean 83.3)
Normalized entries
0.10
0.08
0.06
0.04
0.02
Ratio to parton
1.25
1.00
0.75
0
100
200
300
400
b-quark / b-jet p_T [GeV]

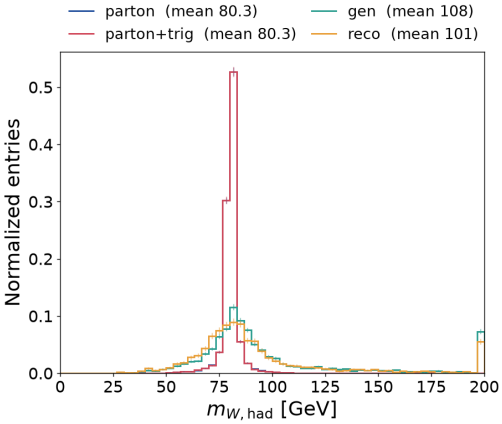
- $\chi^2 = \left(\frac{m_{jj}-m_W}{\sigma_W}\right)^2 + \left(\frac{m_{jjb}-m_t}{\sigma_t}\right)^2 + \left(\frac{m_{\ell\nu b}-m_t}{\sigma_t}\right)^2$ over assignments of the 6 leading jets
- b slots hold $\min(n_b, 2)$ b-tagged jets
- p_z^ν from the m_W constraint; both solutions enter the χ^2, so the m_t term resolves the two-fold ambiguity
- All four jets correct in 23% (gen) and 18% (reco) of reconstructed events

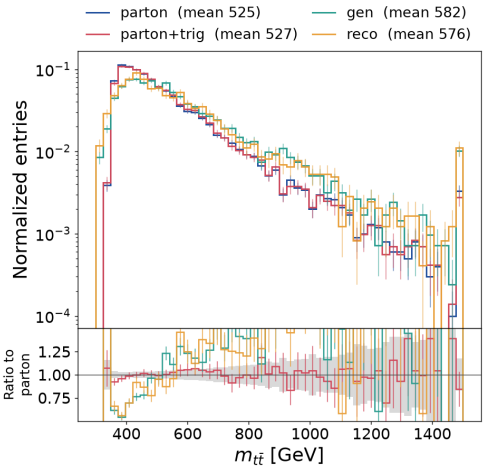
Reconstructed masses (1/2)





Reconstructed masses (2/2)





Mass resolution (median $\pm$ 68% half-width, GeV)

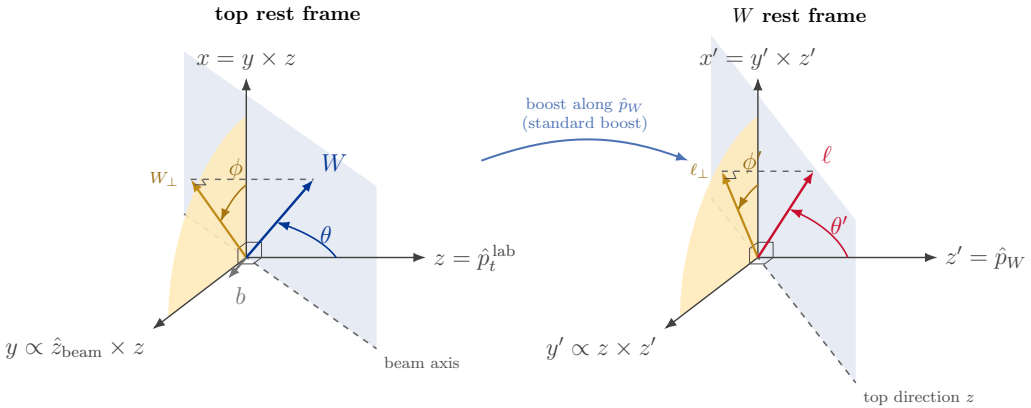
	$m_{t,\mathrm{had}}$, all	$m_{t,\mathrm{had}}$, correct jets	$m_{W,\mathrm{had}}$, all	$m_{W,\mathrm{had}}$, correct jets
gen	182 ± 47	175 ± 14	85 ± 29	83 ± 9
reco	177 ± 44	170 ± 16	84 ± 25	80 ± 10

The broad inclusive peaks come from wrong jet choices (combinatorics) and quarks that are out of acceptance, not from the jet response.

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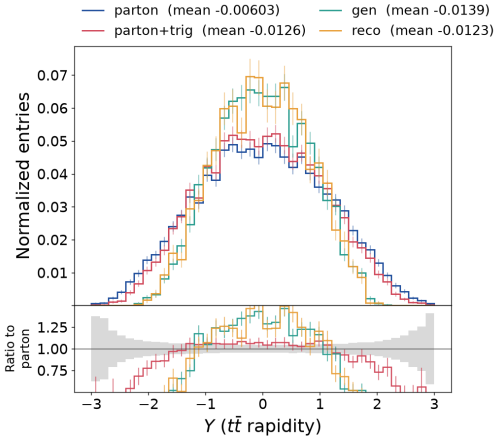
Definition of the ten angles

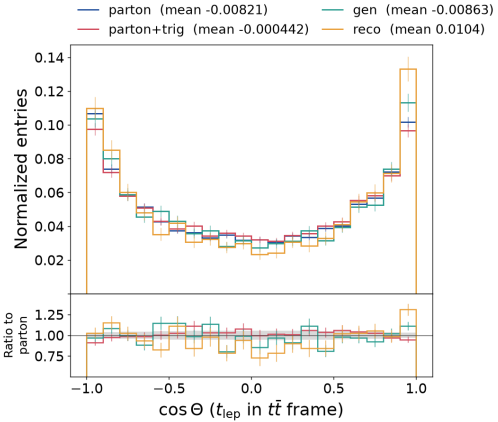


The ten angles (as in the paper)

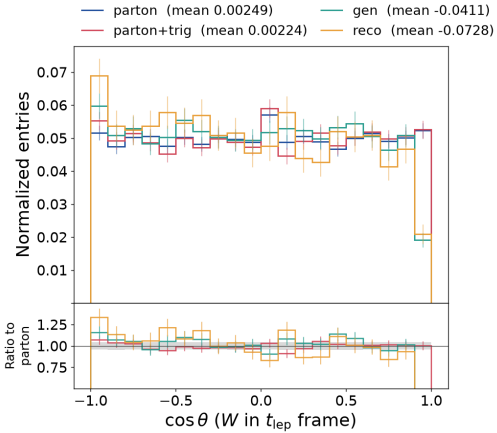
- Y: $t\bar{t}$ rapidity (lab); Θ: polar angle of t_{lep} in the $t\bar{t}$ frame
- (θ, ϕ): W in the top rest frame; $z = \hat{p}_t^{\text{lab}}$, $y = \hat{z}_{\text{beam}} \times z$, $x = y \times z$
- (θ', ϕ'): fermion in the W rest frame (boost from top frame); $z' = \hat{p}_W$, $y' = z \times z'$, $x' = y' \times z'$, i.e. $D(\phi, \theta, 0)$
- Leptonic top: fermion $= \ell$. Hadronic top ($\sim$): fermion = down-type quark
- gen/reco: down-type jet chosen by truth-matching to the LHE quarks (assumes up/down tagging, as in the paper)
- Check: building momenta from known angles and recovering them agrees to 10^{-14}

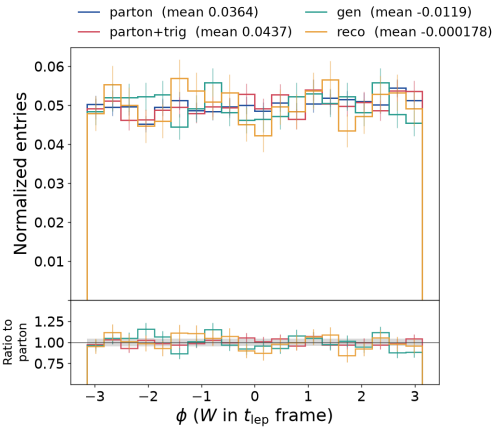
Production angles Y, $\cos\Theta$



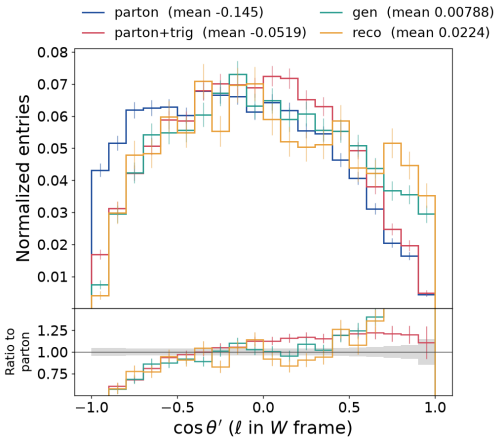


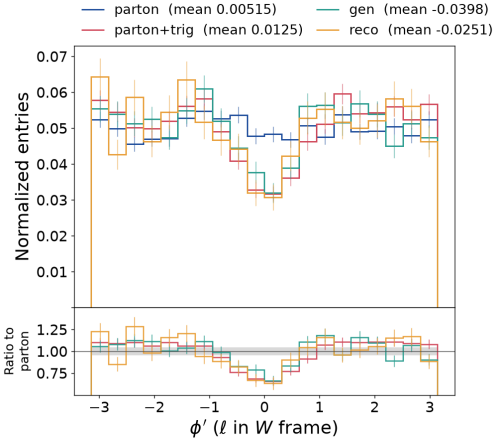
Leptonic top: $\theta, \phi, \theta', \phi'$ (1/2)



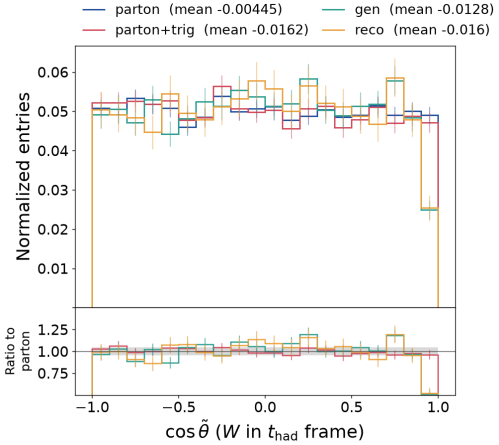


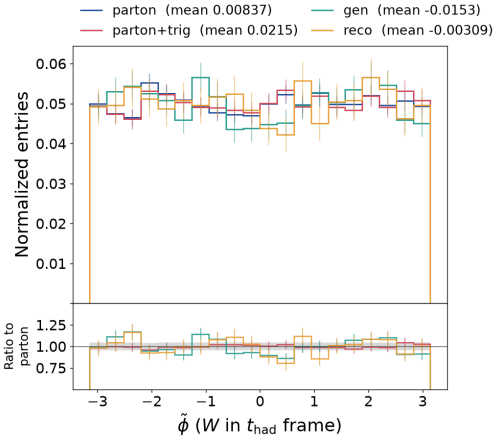
Leptonic top: $\theta, \phi, \theta', \phi'$ (2/2)



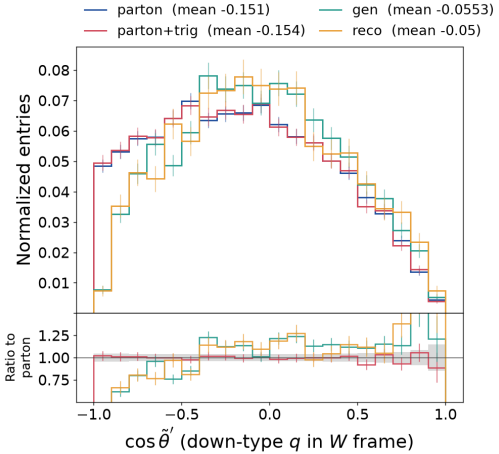


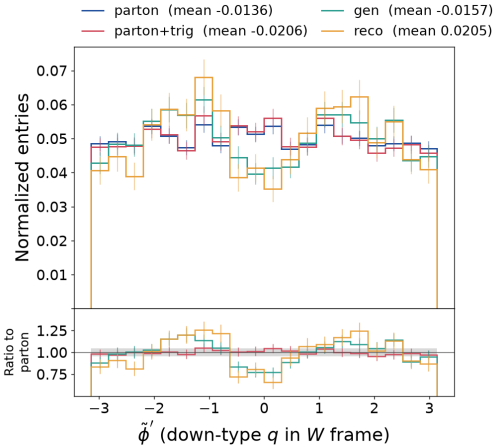
Hadronic top: $\tilde{\theta}, \tilde{\phi}, \tilde{\theta}', \tilde{\phi}'$ (1/2)





Hadronic top: $\tilde{\theta}, \tilde{\phi}, \tilde{\theta}', \tilde{\phi}'$ (2/2)





W helicity fractions from $\cos\theta'$ (moments)

	F_0	F_L	F_R
parton, ℓ	0.715 ± 0.013	0.288 ± 0.008	-0.003 ± 0.008
parton, down-type q	0.705 ± 0.013	0.298 ± 0.008	-0.003 ± 0.008
gen, ℓ	0.769 ± 0.019	0.107 ± 0.012	0.123 ± 0.012
gen, down-type q	0.954 ± 0.017	0.078 ± 0.011	-0.032 ± 0.011
reco, ℓ	0.687 ± 0.026	0.134 ± 0.017	0.179 ± 0.017
reco, down-type q	0.929 ± 0.023	0.086 ± 0.015	-0.014 ± 0.015

SM (LO): $F_0 = m_t^2/(m_t^2 + 2m_W^2) \approx 0.70$, $F_L \approx 0.30$, $F_R \approx 0$. Parton level reproduces it; gen/reco are distorted by acceptance and combinatorics (no unfolding).

Angle resolution vs parton level (correlation / median $|\Delta|$)

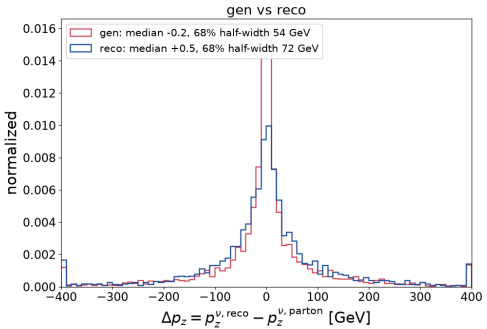
angle	gen, all	gen, correct jets	reco, all	reco, correct jets
Y	0.90 / 0.16	0.99 / 0.05	0.89 / 0.18	0.98 / 0.06
$\cos\Theta$	0.67 / 0.13	0.97 / 0.04	0.64 / 0.15	0.97 / 0.06
$\cos\theta$	0.70 / 0.14	0.89 / 0.06	0.62 / 0.22	0.84 / 0.11
ϕ	0.48 / 0.36	0.82 / 0.12	0.40 / 0.52	0.89 / 0.16
$\cos\theta'$	0.68 / 0.11	0.92 / 0.04	0.60 / 0.20	0.89 / 0.11
ϕ'	0.45 / 0.37	0.68 / 0.13	0.30 / 0.63	0.47 / 0.28
$\cos\tilde{\theta}$	0.59 / 0.18	0.98 / 0.04	0.45 / 0.24	0.97 / 0.05
$\tilde{\phi}$	0.40 / 0.46	0.88 / 0.09	0.34 / 0.69	0.90 / 0.09
$\cos\tilde{\theta}'$	0.41 / 0.18	0.97 / 0.04	0.31 / 0.25	0.96 / 0.06
$\tilde{\phi}'$	0.39 / 0.55	0.93 / 0.06	0.33 / 0.73	0.94 / 0.07

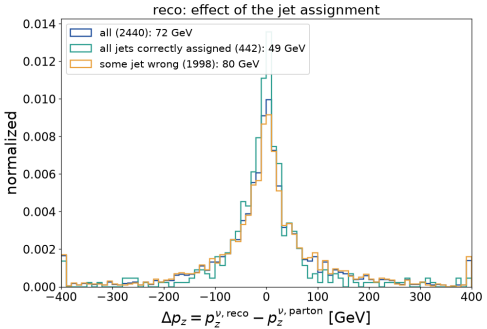
Neutrino p_z resolution

	median Δp_z [GeV]	68% half-width [GeV]
gen	-0.25	54
reco	0.5	72.2

$\Delta p_z = p_z^{\nu,\mathrm{reco}} - p_z^{\nu,\mathrm{parton}}$, events with a reconstructed $t\bar{t}$ system.

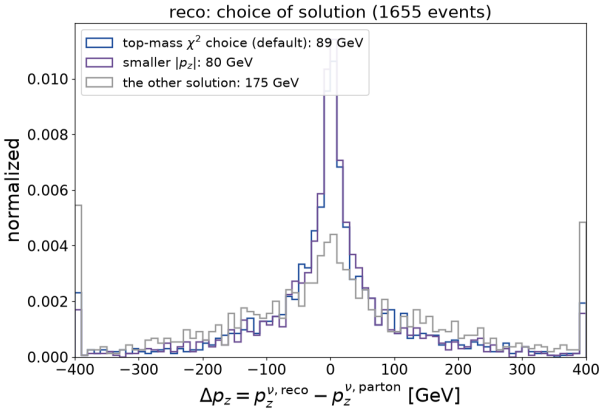
Neutrino Δp_z





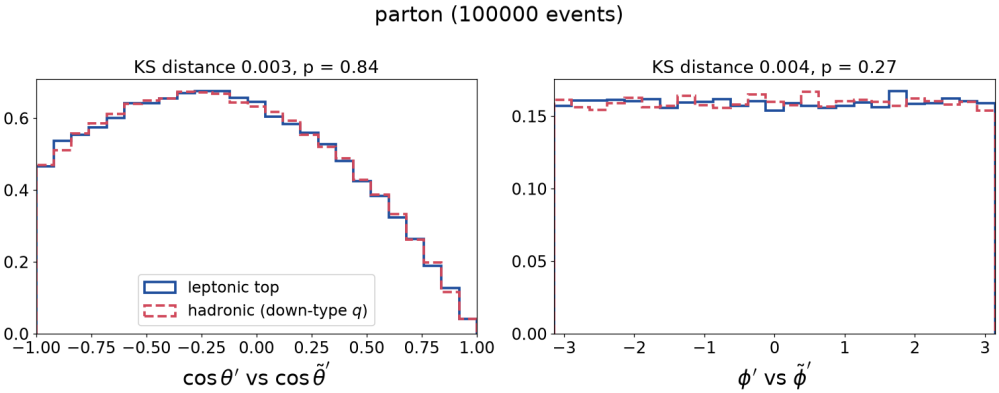
68% half-width: gen 54 GeV, reco 72 GeV; reco with all jets correct 49 GeV.

Neutrino Δp_z: which m_W solution?



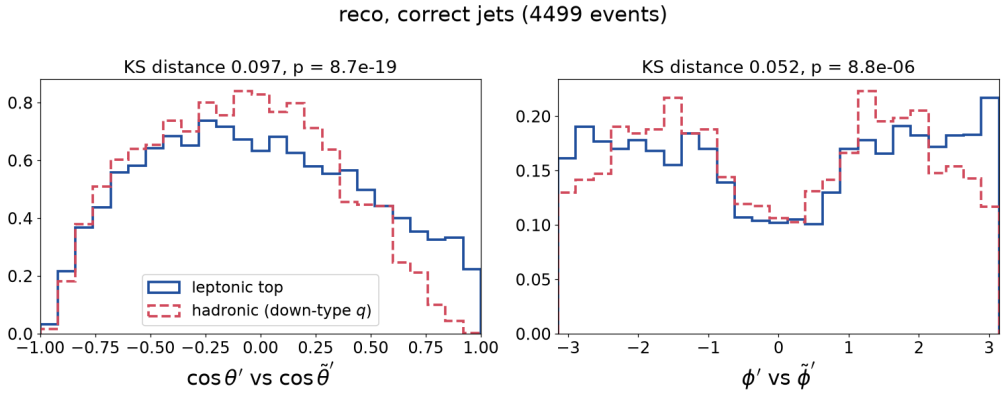
No real solution in 32% of events. With two, the top-mass χ^2 picks the one closer to the truth in 63% of events vs 65% for the smaller $|p_z|$.

Leptonic vs hadronic top, parton level: W decay angles



SM: the down-type quark plays the role of the charged lepton. Parton level agrees (KS $p = 0.84$, 0.27, 0.24, 0.54 for $\cos\theta$, ϕ, $\cos\theta'$, ϕ').

Leptonic vs hadronic top, reco level with correct jets



Even with perfect flavour tagging and correct jets they differ ($\cos\theta'$ KS distance 0.097): different acceptance and resolution on the two sides (lepton cuts + ν vs jet cuts and response).

Study A: $t\bar{t}$ vs $hh \to b\bar{b}WW^*$ (an off-shell signal)

- $f_{t\bar{t}}$ on the ten angles: fits, coefficients, bases, acceptance
- Why the separation washes out after reconstruction, and the fix: the resonance masses
- Comparison with other unsupervised methods and with supervised ML on the same inputs

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- MadGraph5 built-in `heft` model (LO, $m_t \to \infty$), same Pythia8 + Delphes ATLAS chain
- $gg \to hh$, $h \to b\bar{b}$, $h \to \ell\nu q\bar{q}'$ (+ charge conjugate), generated as one process
- $h \to WW^*$ has one off-shell W, so $h \to \ell\nu q\bar{q}'$ is a 4-body decay
- 20000 events, seed 2001; $\sigma \times \mathcal{B} = 0.112$ fb (LO HEFT with MG defaults; *not* an SM prediction, so plots use unit area)
- For $t\bar{t}$-hypothesis quantities, hh uses the same χ^2; at parton level the b pairing is the one closest to m_t
- Prompt leptons: status-1 e/μ from W, Z, h, or a τ from one of these (in $h \to WW^{(*)}$ the lepton can come directly from the h in the record)

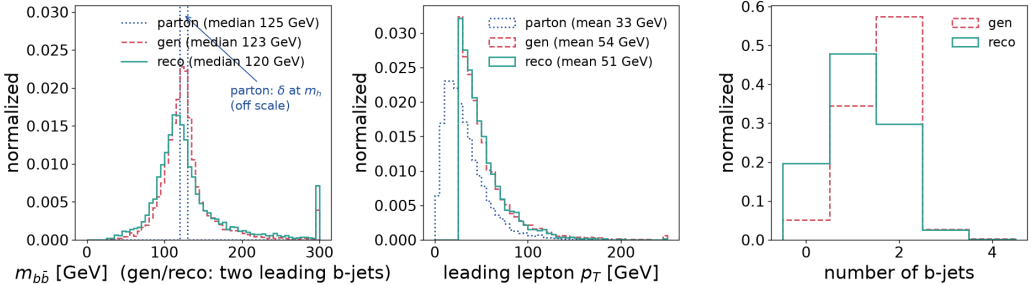
Cutflow: $t\bar{t}$ vs hh (reco)

selection	$t\bar{t}$		hh	
All events	10000	100.0%	20000	100.0%
exactly 1 lepton	4907	49.1%	5403	27.0%
≥ 4 jets	2830	28.3%	2133	10.7%
≥ 1 b-jet	2454	24.5%	1887	9.4%
≥ 2 b-jets	1221	12.2%	920	4.6%
χ^2 reconstruction (from ≥ 1 b)	2440	24.4%	1878	9.4%

hh leptons are softer (from W^*) and less isolated, so the efficiency is lower.

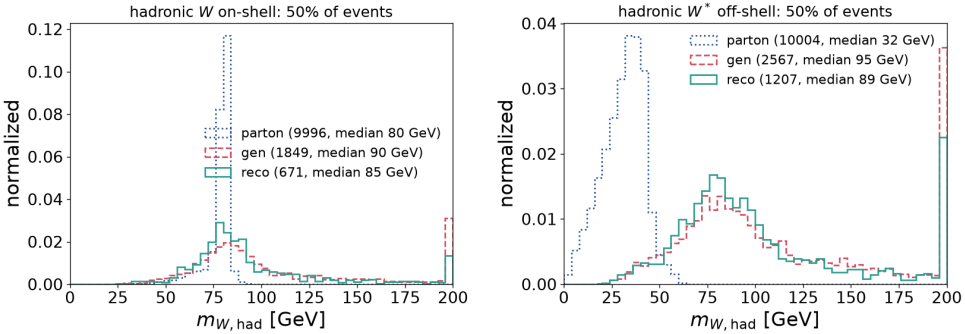
hh sample: parton, gen and reco level

$hh \to b\bar{b}\,WW^* \to b\bar{b}\,\ell\nu\,q\bar{q}'$ at parton, gen and reco level



hh: hadronic W mass, on-shell W vs W^*

$hh \to b\bar{b}\,WW^*$: hadronic W mass by case (gen/reco: χ^2-selected jets)



Hadronic boson is the W^* in 50% of events (true median 32 GeV), but the $t\bar{t}$-hypothesis χ^2 picks jet pairs near m_W (median 95 / 89 GeV at gen / reco): the reconstruction makes hh look like $t\bar{t}$.

Outline

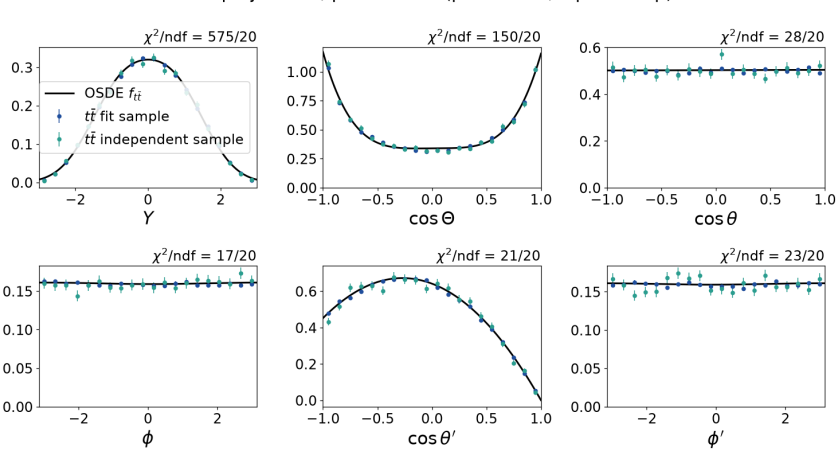
1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Spherical-harmonic basis
9. Is the OSDE approach helping? (ML)
10. Why the separation washes out at gen level
11. Adding momentum to y: the resonance masses
12. Discriminator t (paper style)
13. Acceptance
14. Study A′: hh with on-shell W bosons ($m_H = 170$ GeV)
15. Study B: $t\bar{t}$ vs $H \to t\bar{t}$
16. Pileup
17. Level-1 trigger: a lookup-table version of angles + BW
18. Summary

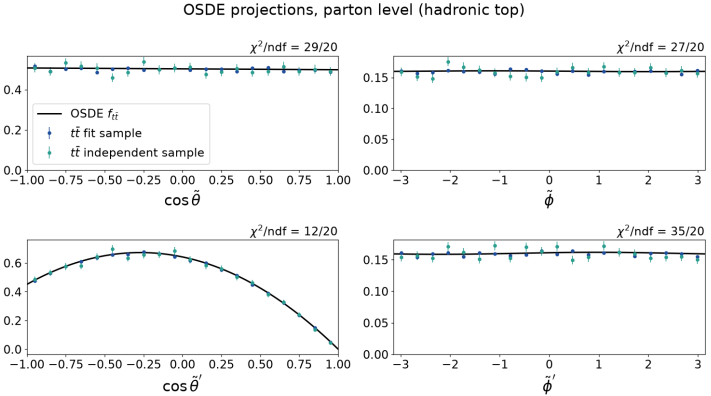
Basis functions (as in the paper)

- $M^{j_1 j_2}_{m' m}(\phi, \theta, \phi', \theta') = \frac{\sqrt{2j_1+1}\sqrt{2j_2+1}}{4\pi} D^{j_1}_{m' m}(\phi, \theta, 0)\, D^{j_2}_{m 0}(\phi', \theta', 0)$, $j_1 \leq 1$, $j_2 \leq 2$: 24 per top
- $p_l(\cos\Theta) = \sqrt{\frac{2l+1}{2}} P_l$, $h_n(Y) = (2^n n! \sqrt{\pi})^{-1/2} H_n(Y) e^{-Y^2/2}$, $l < 5$, $n < 5$
- $B = M_{\mathbf{j}} M_{\tilde{\mathbf{j}}}\, p_l\, h_n$: $24 \times 24 \times 5 \times 5 = 14400$ coefficients
- OSDE: $\hat{c} = \frac{1}{N} \sum_k B^*(\mathbf{x}_k)$, with standard error; $f_{t\bar{t}} = \sum \hat{c}\, B$ (real, since the set is closed under conjugation)
- Hermite *functions*: bare H_n are orthogonal only with weight e^{-Y^2}
- **Three separate fits**: $f^{\text{parton}}_{t\bar{t}}$ from parton-level $t\bar{t}$ angles, applied to parton-level $t\bar{t}$ and hh; likewise f^{gen} (gen on gen) and f^{reco} (reco on reco)

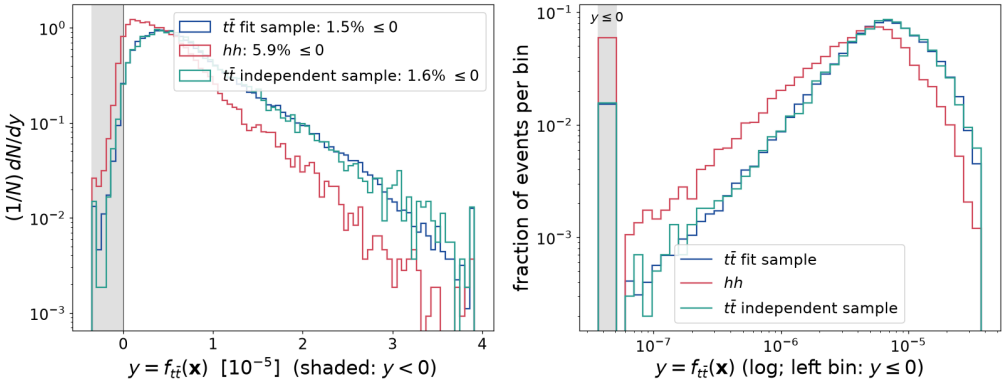
Checks of the basis and of OSDE

- Unit tests: not run
- Wigner $d^{j}_{m'm}$ agrees with closed forms and with `sympy` to 10^{-12}
- Gram matrices of M (exact quadrature), p_l, h_n equal the identity to 10^{-12}
- Toy: 2×10^5 events from a density with known coefficients; all 14400 recovered with pulls of unit width; projections and $\int f = 1$ reproduced
- Fitted f: imaginary part $< 10^{-10}$ of the real part
- Extended basis: Wigner d stable to $j = 24$ (Jacobi form); blocks disjoint and orthonormal; reduces exactly to the paper basis; leave-one-out equals brute-force refits

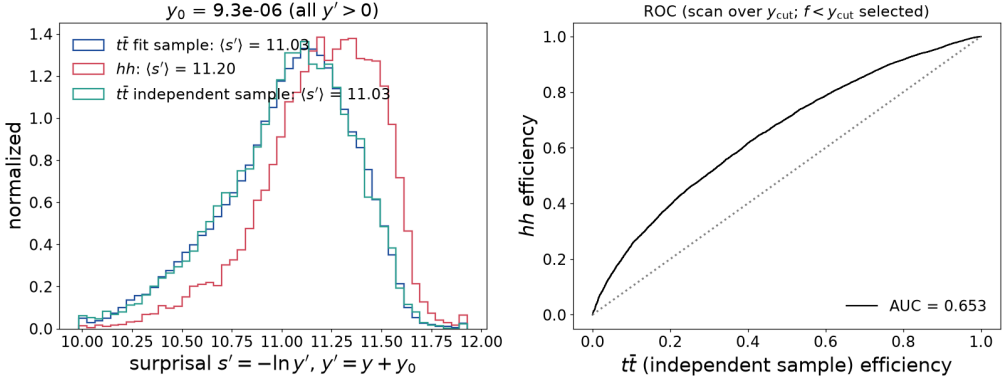

OSDE projections, parton level (production, leptonic top)
χ^2/ndf = 575/20
OSDE $f_{t\bar{t}}$
$t\bar{t}$ fit sample
$t\bar{t}$ independent sample
Y
χ^2/ndf = 150/20
$\cos\Theta$
χ^2/ndf = 28/20
$\cos\theta$
χ^2/ndf = 17/20
ϕ
χ^2/ndf = 21/20
$\cos\theta'$
χ^2/ndf = 23/20
ϕ'



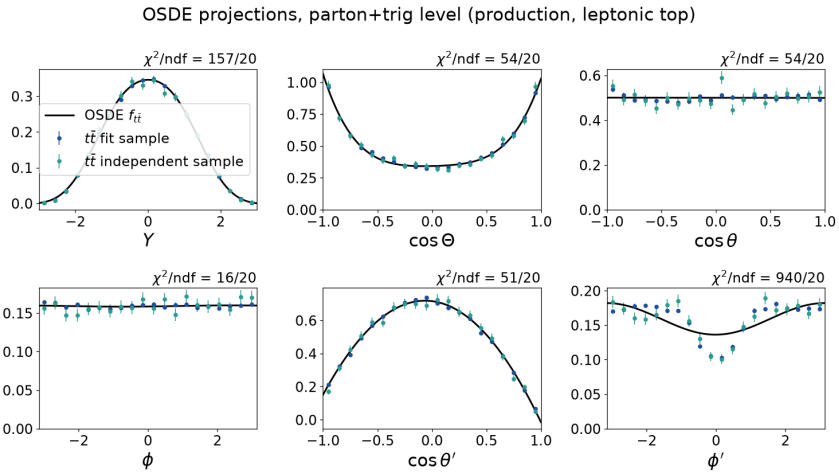
Black: $f_{t\bar{t}}$ integrated over the other nine angles. Largest χ^2/ndf vs training data: Y (575/20).

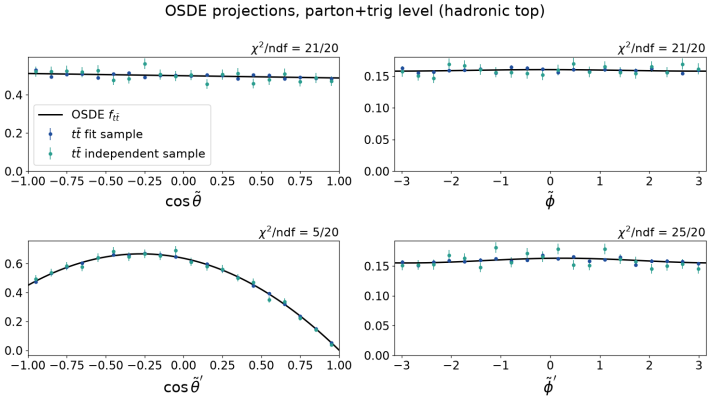


Left: $y = f_{t\bar{t}}(\mathbf{x})$, linear (shaded: $y < 0$). Right: log scale; the left bin holds all events with $y \leq 0$. $t\bar{t}$: independent sample and fit sample (leave-one-out).

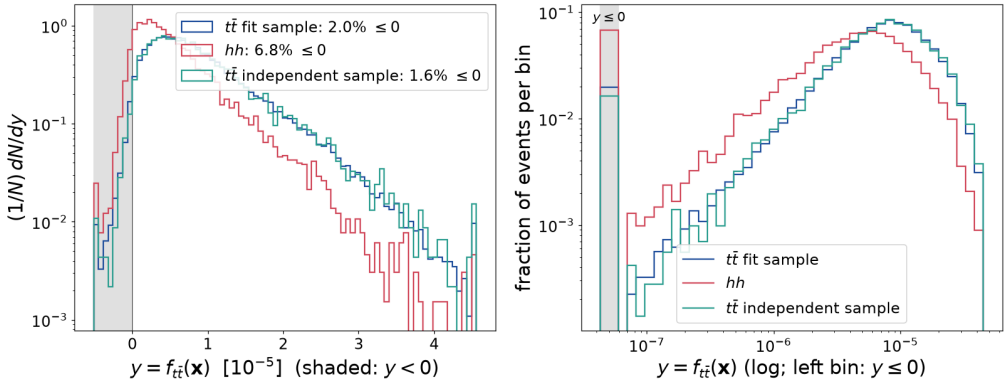


Left: $s' = -\ln(y + y_0)$, shifted so every event has $y' > 0$. Right: ROC from a scan over y_{cut} (events with $f < y_{\text{cut}}$ called hh-like).

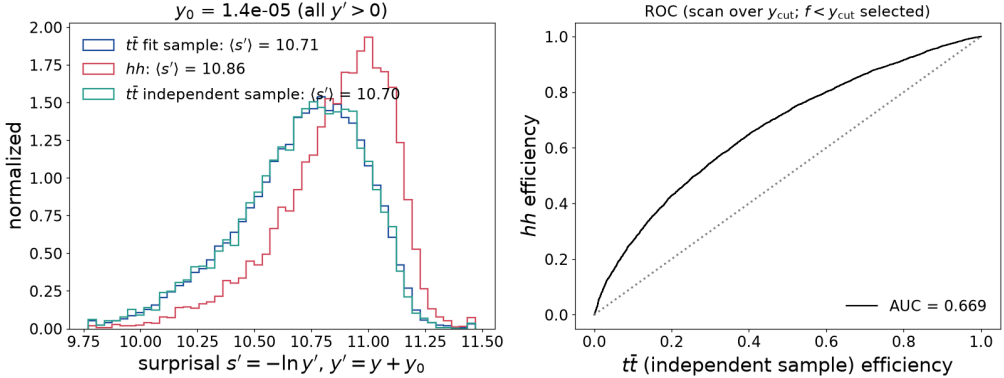

OSDE projections, parton+trig level (production, leptonic top)
χ²/ndf = 157/20
OSDE $f_{t\bar{t}}$
$t\bar{t}$ fit sample
$t\bar{t}$ independent sample
Y
χ²/ndf = 54/20
cos Θ
χ²/ndf = 54/20
cos θ
χ²/ndf = 16/20
ϕ
χ²/ndf = 51/20
cos θ′
χ²/ndf = 940/20
ϕ′



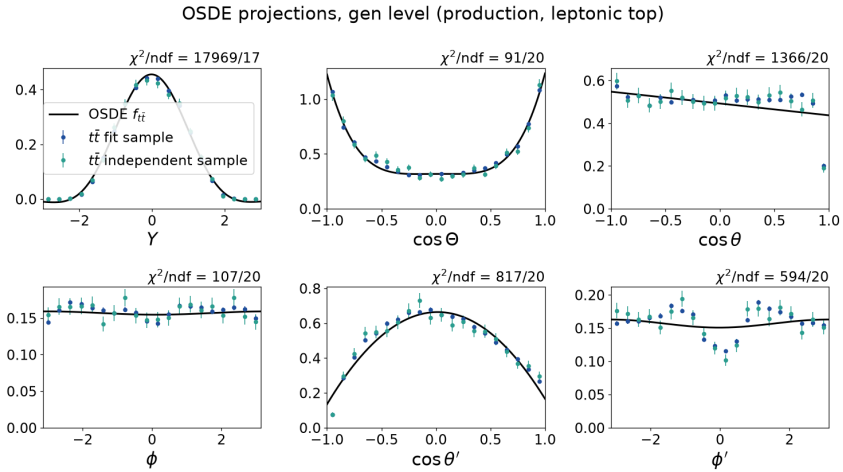
Black: $f_{t\bar{t}}$ integrated over the other nine angles. Largest χ^2/ndf vs training data: ϕ' (940/20).

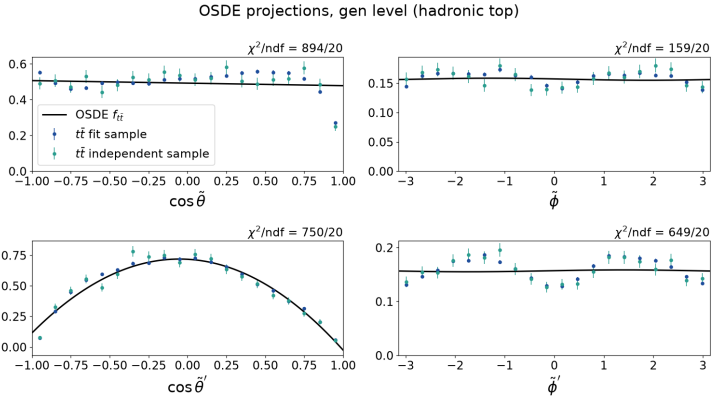


Left: $y = f_{t\bar{t}}(\mathbf{x})$, linear (shaded: $y < 0$). Right: log scale; the left bin holds all events with $y \leq 0$. $t\bar{t}$: independent sample and fit sample (leave-one-out).

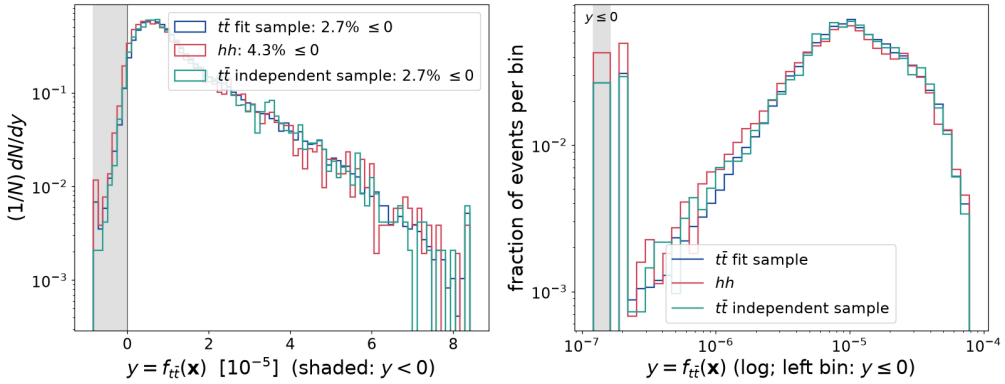


Left: $s' = -\ln(y + y_0)$, shifted so every event has $y' > 0$. Right: ROC from a scan over y_{cut} (events with $f < y_{\text{cut}}$ called hh-like).

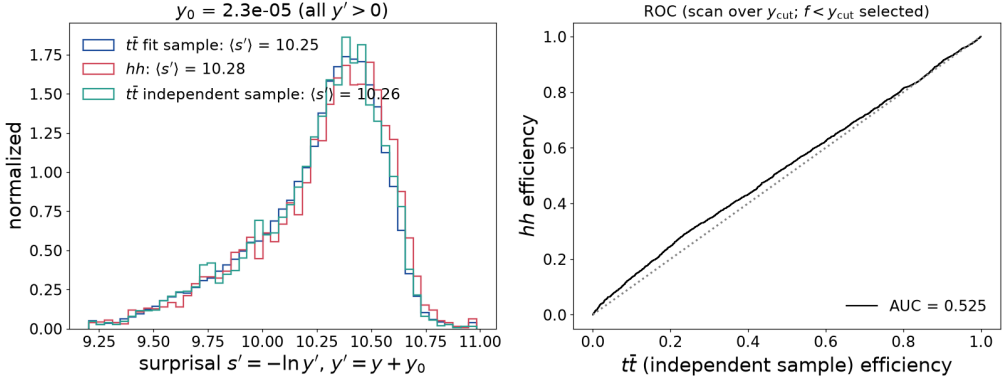

OSDE projections, gen level (production, leptonic top)
χ^2/ndf = 17969/17
OSDE $f_{t\bar{t}}$
$t\bar{t}$ fit sample
$t\bar{t}$ independent sample
Y
χ^2/ndf = 91/20
$\cos\Theta$
χ^2/ndf = 1366/20
$\cos\theta$
χ^2/ndf = 107/20
ϕ
χ^2/ndf = 817/20
$\cos\theta'$
χ^2/ndf = 594/20
ϕ'



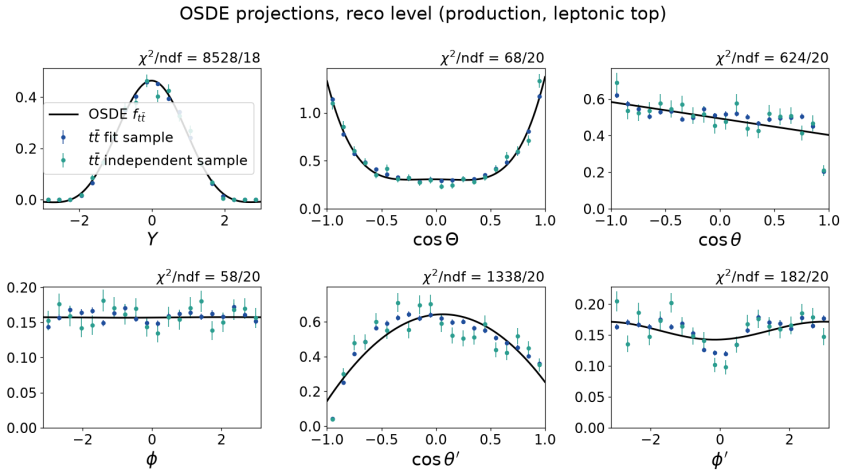
Black: $f_{t\bar{t}}$ integrated over the other nine angles. Largest χ^2/ndf vs training data: Y (17969/17).

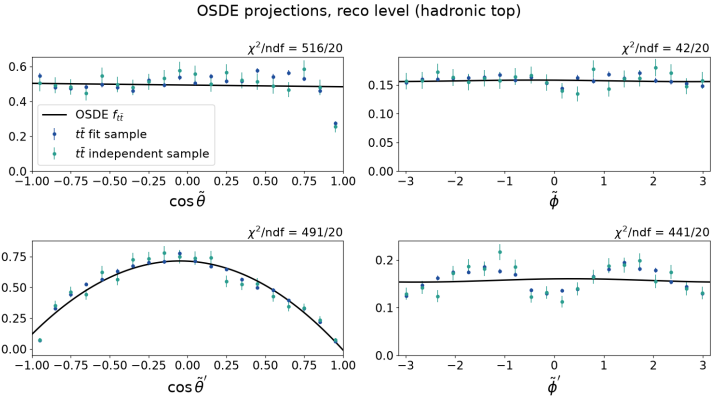


Left: $y = f_{t\bar{t}}(\mathbf{x})$, linear (shaded: $y < 0$). Right: log scale; the left bin holds all events with $y \leq 0$. $t\bar{t}$: independent sample and fit sample (leave-one-out).

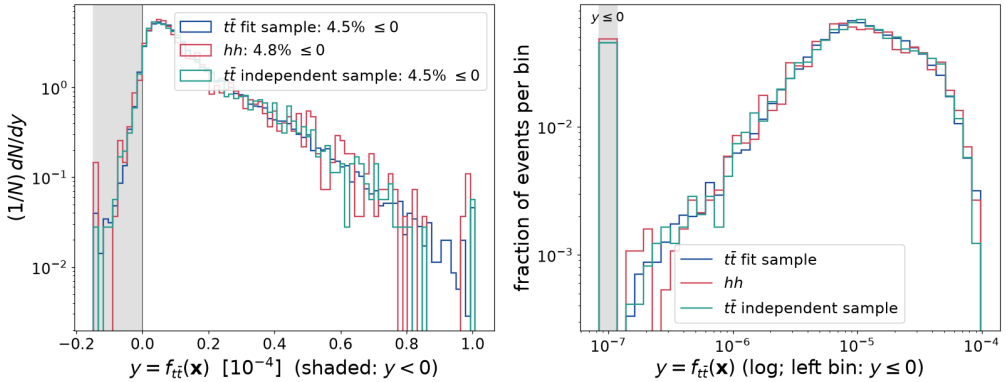


Left: $s' = -\ln(y + y_0)$, shifted so every event has $y' > 0$. Right: ROC from a scan over y_{cut} (events with $f < y_{\text{cut}}$ called hh-like).

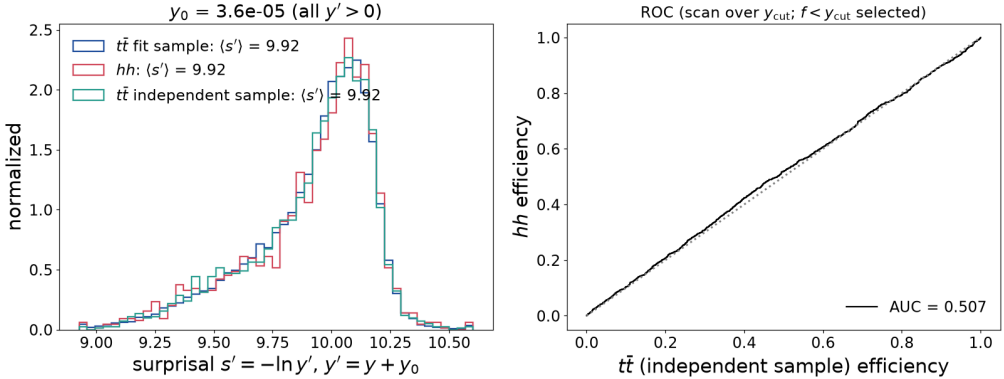

OSDE projections, reco level (production, leptonic top)
χ^2/ndf = 8528/18
OSDE $f_{t\bar{t}}$
$t\bar{t}$ fit sample
$t\bar{t}$ independent sample
Y
χ^2/ndf = 68/20
$\cos\Theta$
χ^2/ndf = 624/20
$\cos\theta$
χ^2/ndf = 58/20
ϕ
χ^2/ndf = 1338/20
$\cos\theta'$
χ^2/ndf = 182/20
ϕ'



Black: $f_{t\bar{t}}$ integrated over the other nine angles. Largest χ^2/ndf vs training data: Y (8528/18).



Left: $y = f_{t\bar{t}}(\mathbf{x})$, linear (shaded: $y < 0$). Right: log scale; the left bin holds all events with $y \leq 0$. $t\bar{t}$: independent sample and fit sample (leave-one-out).



Left: $s' = -\ln(y + y_0)$, shifted so every event has $y' > 0$. Right: ROC from a scan over y_{cut} (events with $f < y_{\text{cut}}$ called hh-like).

level	N_{train}	$\lvert\hat{c}\rvert > 3\sigma$	$\int f$	$f \leq 0$: $t\bar{t}/hh$	$\langle s \rangle$: $t\bar{t}/hh$	AUC
parton	100000	186	1.006	1.6% / 5.9%	12.07 / 12.56	0.653
parton+trig	71322	255	1.000	1.6% / 6.8%	11.90 / 12.45	0.669
gen	41317	330	0.985	2.7% / 4.3%	11.59 / 11.66	0.525
reco	24030	253	0.987	4.5% / 4.8%	11.53 / 11.56	0.507

$t\bar{t}$ numbers are from the independent test sample. $f \leq 0$ comes from the truncated series (Gibbs overshoot), as noted in the paper. AUC = probability that an hh event has lower $f_{t\bar{t}}$ than a $t\bar{t}$ event.

Fit sample vs independent sample: self-inclusion bias

level	self-term / f	$f \leq 0$: fit / fit LOO / indep.	$\langle s \rangle$: fit / fit LOO / indep.	KS vs indep.: fit / LOO
parton	7%	0.6% / 1.5% / 1.6%	11.98 / 12.08 / 12.07	0.041 / 0.010
parton+trig	8%	0.7% / 2.0% / 1.6%	11.81 / 11.91 / 11.90	0.048 / 0.008
gen	10%	1.0% / 2.7% / 2.7%	11.44 / 11.57 / 11.59	0.071 / 0.012
reco	16%	1.2% / 4.5% / 4.5%	11.33 / 11.53 / 11.53	0.103 / 0.012

Each fit event adds $\frac{1}{N}\sum_j |B_j(\mathbf{x}_k)|^2$ to its own f. Removing it (leave-one-out, LOO) makes the fit sample agree with the independent $t\bar{t}$ sample: the difference was entirely this bias, largest at reco (smallest N). Plots use LOO for the fit sample.

Diagnostics: statistical noise of the 14400 coefficients

level	all (paper)	$\lvert\hat{c}\rvert > 2\sigma$	$\lvert\hat{c}\rvert > 3\sigma$	$\lvert\hat{c}\rvert > 5\sigma$
parton	14400 / 0.653 / $+0.089$	751 / 0.663 / $+0.014$	186 / 0.666 / $+0.001$	54 / 0.669 / -0.005
parton+trig	14400 / 0.669 / $+0.096$	785 / 0.683 / $+0.006$	255 / 0.689 / -0.006	108 / 0.691 / -0.014
gen	14400 / 0.525 / $+0.144$	827 / 0.525 / $+0.042$	330 / 0.525 / $+0.025$	153 / 0.527 / $+0.009$
reco	14400 / 0.507 / $+0.202$	712 / 0.507 / $+0.040$	253 / 0.504 / $+0.025$	127 / 0.507 / $+0.015$

Entries: coefficients kept / AUC(hh vs $t\bar{t}$ test) / $\langle s \rangle_{\mathrm{test}} - \langle s \rangle_{\mathrm{train}}$ (overfitting). Dropping insignificant terms removes the overfitting, but the gen/reco AUC stays ≈ 0.5, so the loss of separation is not caused by noise.

Diagnostics: Hermite scale for Y (χ^2 of the Y projection, train)

level	$s = 1.0$	$s = 0.9$	$s = 0.8$	$s = 0.7$	$s = \mathrm{std}(Y)$
parton	660/24	998/24	379/24	507/24	230/24
parton+trig	172/24	904/24	1013/24	117/24	111/24
gen	2298/20	2956/20	548/20	728/20	548/20
reco	4485/21	3864/21	721/21	361/21	515/21

The paper uses $s = 1$ (the default here; it is a free parameter of the fit). Gen/reco Y is narrower (std ≈ 0.8), so $s = 1$ fits poorly there and f overshoots below zero at large $|Y|$.

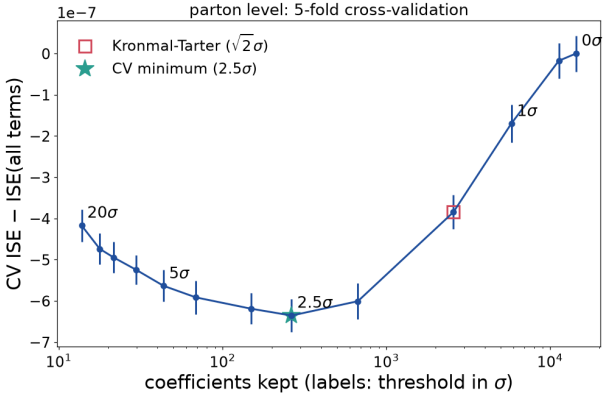
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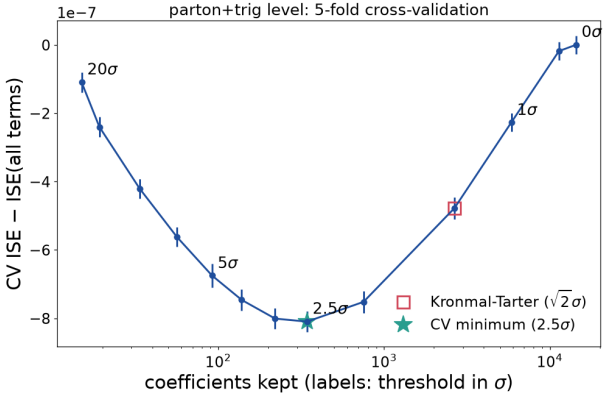
Criterion for dropping coefficients

- Keep a term iff $|\hat{c}|/\sigma(\hat{c}) > t$ (constant always kept). Conjugate partners have equal $|\hat{c}|$ and σ, so they are kept together and f stays real (checked)
- Kronmal–Tarter ($t = \sqrt{2}$): dropping a term costs $|c|^2$, keeping it costs σ^2; keep iff $|\hat{c}|^2 > 2\sigma^2$ (minimizes the integrated squared error per term)
- With 14400 mostly-null terms, $\sqrt{2}\sigma$ still admits $\sim$16% of the pure-noise terms
- Choose t by 5-fold cross-validation: for an orthonormal basis, $\mathrm{ISE} - \int f^2 = \sum_{\text{kept}} |\hat{c}|^2 - 2\,\mathrm{Re} \sum_{\text{kept}} \hat{c}\,\hat{c}^*_{\text{held-out}}$ (no integrals needed)
- CV minimum at $t = 2.5$: 326 (parton), 385 (parton+trig), 459 (gen), 342 (reco) of 14400 terms; flat out to 5σ: 54, 108, 153, 127

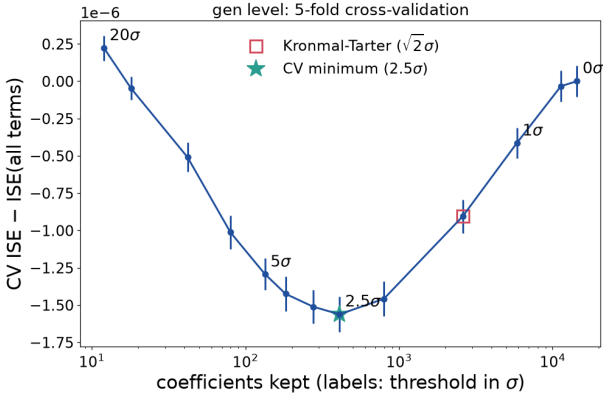
Cross-validated error vs number of coefficients, parton



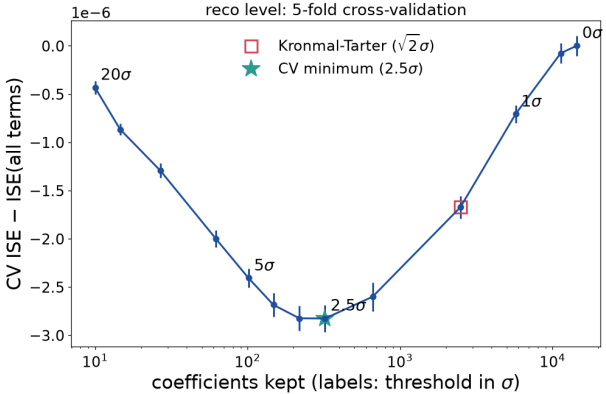
Cross-validated error vs number of coefficients, parton+trig



Cross-validated error vs number of coefficients, gen



Cross-validated error vs number of coefficients, reco



Simplified $f_{t\bar{t}}$: performance (1/2)

level	terms kept	number	AUC	$f \leq 0$ ($t\bar{t}$ indep.)	$f \leq 0$ (hh)	$\langle s \rangle$ fit LOO / indep.
parton	all (paper)	14400	0.653	1.6%	5.9%	12.08 / 12.07
	$\sqrt{2}\sigma$ (K–T)	2629	0.658	1.0%	4.8%	12.04 / 12.06
	CV best	326	0.665	0.2%	2.4%	12.06 / 12.06
	5σ	54	0.669	0.1%	2.1%	12.07 / 12.06
parton+trig	all (paper)	14400	0.669	1.6%	6.8%	11.91 / 11.90
	$\sqrt{2}\sigma$ (K–T)	2756	0.677	1.2%	5.7%	11.87 / 11.89
	CV best	385	0.685	0.4%	3.4%	11.91 / 11.89
	5σ	108	0.691	0.4%	3.7%	11.92 / 11.90
gen	all (paper)	14400	0.525	2.7%	4.3%	11.57 / 11.59
	$\sqrt{2}\sigma$ (K–T)	2725	0.526	1.9%	2.9%	11.53 / 11.57
	CV best	459	0.525	0.9%	1.3%	11.56 / 11.58
	5σ	153	0.527	1.0%	1.5%	11.57 / 11.58

Simplified $f_{t\bar{t}}$: performance (2/2)

level	terms kept	number	AUC	$f \leq 0$ ($t\bar{t}$ indep.)	$f \leq 0$ (hh)	$\langle s \rangle$ fit LOO / indep.
reco	all (paper)	14400	0.507	4.5%	4.8%	11.53 / 11.53
	$\sqrt{2}\sigma$ (K–T)	2551	0.508	2.7%	2.8%	11.47 / 11.52
	CV best	342	0.505	1.1%	1.5%	11.52 / 11.54
	5σ	127	0.507	1.5%	1.9%	11.54 / 11.55

Dropping insignificant terms cuts the negative-f (Gibbs) fraction and keeps the hh separation: the simplified f is as good, with $\sim$1–3% of the terms.

Leading terms at parton level (after the constant)

#	term	$\lvert\hat{c}\rvert/\sigma$	physics
1	$P_2(\cos\Theta)$	115	production angle
2	$M_{0,0}^{01}$	88	W helicity (lep)
3	$\tilde{M}_{0,0}^{01}$	87	W helicity (had)
4	$M_{0,0}^{02}$	80	W helicity (lep)
5	$\tilde{M}_{0,0}^{02}$	79	W helicity (had)
6	$h_2(Y)$	59	rapidity
7	$P_4(\cos\Theta)$	34	production angle
8	$M_{0,0}^{01} \times P_2(\cos\Theta)$	32	W helicity (lep); production angle

M: leptonic top, $\tilde{M}$: hadronic top; $j_1 = 0$: W helicity, $j_1 = 1$: top spin. Production (P_l), W helicity and Y dominate.

What the significant ($> 5\sigma$) terms describe

physics content of the term	parton	parton+trig	gen	reco
production angle + top pol. $\times W$ hel. interference	–	10	30	26
rapidity + top pol. $\times W$ hel. interference	–	15	24	12
top pol. $\times W$ hel. interference + $t\bar{t}$ spin correlation	16	12	–	–
W helicity + production angle	12	7	8	6
production angle + rapidity + top polarization	–	1	12	12
production angle + rapidity + top pol. $\times W$ hel. interference	–	7	11	11
W helicity + production angle + rapidity	1	2	11	11
W helicity + top pol. $\times W$ hel. interference	–	10	4	4
top pol. $\times W$ hel. interference + top polarization + $t\bar{t}$ spin correlation	6	10	–	–
W helicity	8	8	6	6
total	**53**	**107**	**152**	**126**

Parton: production, W helicity and $t\bar{t}$ spin correlations; imaginary parts consistent with 0 (max 1.4σ), as expected without CP violation. Reco: spin correlations no longer resolved; acceptance-induced "polarization" terms appear instead.

Outline

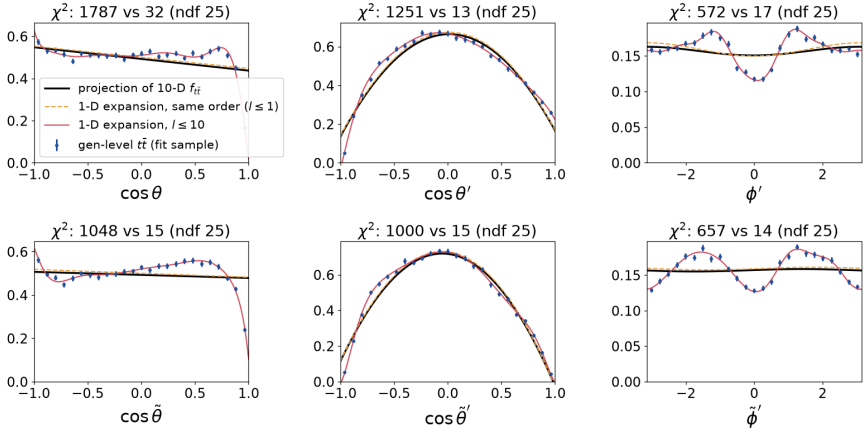
1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Spherical-harmonic basis
9. Is the OSDE approach helping? (ML)
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16. Pileup
17. Level-1 trigger: a lookup-table version of angles + BW
18. Summary

Why f^{gen}, f^{reco} miss the data with the paper basis

- $j_1 \leq 1$: at most *linear* in $\cos\theta$; $j_2 \leq 2$: at most *quadratic* in $\cos\theta'$; $|m| \leq 1$: only $\cos\phi'$, $\sin\phi'$
- Complete for the physical (parton) decay distribution, but the acceptance cuts create sharp edges ($\cos\theta \to 1$, $\cos\theta' \to -1$, $\phi' \approx 0$) that need higher orders
- Code check: the projection of f equals a direct 1-D expansion at the same order (max difference 1.9%)
- At gen level, same-order χ^2: 1787, 1251, 572, 1048, 1000, 657 (25 bins); higher order: 32, 13, 17, 15, 15, 14

Paper basis at gen level: the order, not the fit, limits f

gen level: the basis order limits what $f_{t\bar{t}}$ can describe (black = dashed orange: code consistent)



How many orders each angle needs (1-D cross-validated order)

angle	parton	parton+trig	gen	reco
$\cos\theta$	0	4	13	18
ϕ	0	2	4	4
$\cos\theta'$	2	8	6	5
ϕ'	1	7	12	4
$\cos\tilde{\theta}$	1	1	7	7
$\tilde{\phi}$	0	0	4	3
$\cos\tilde{\theta}'$	2	2	8	24
$\tilde{\phi}'$	1	1	4	5
$\cos\Theta$	10	6	6	13
Y	10	10	8	24
needs	$j_1 \geq 1, j_2 \geq 2$	$j_1 \geq 4, j_2 \geq 8$	$j_1 \geq 13, j_2 \geq 12$	$j_1 \geq 18, j_2 \geq 24$

Order = Legendre degree ($\cos$), harmonic (ϕ), Hermite order (Y, scale = std(Y)). Parton level confirms the paper: $j_1 \leq 1$, $j_2 \leq 2$ (only $\cos\Theta$, Y need more). A full product at gen/reco orders would need $> 10^6$ terms per top pair.

Extended (block) basis

- Union of disjoint subsets of the full orthonormal product basis $\Rightarrow$ still orthonormal, OSDE unchanged
- A: the paper's full product (all correlations at physics order); L, H: high-order single-top blocks (all (θ, ϕ) terms, all $m' = 0$ terms) with the other top constant; P, Y: high-order $\cos\Theta$ and Y
- Orders from the scan, capped at 16: parton: $j_1 \leq 1$, $j_2 \leq 2$, 11 Legendre, 11 Hermite ($s = 1.111$); parton+trig: $j_1 \leq 7$, $j_2 \leq 8$, 7 Legendre, 11 Hermite ($s = 1.03$); gen: $j_1 \leq 13$, $j_2 \leq 12$, 7 Legendre, 9 Hermite ($s = 0.799$); reco: $j_1 \leq 16$, $j_2 \leq 16$, 14 Legendre, 17 Hermite ($s = 0.791$)
- Terms: parton 14436, parton+trig 18936, gen 32494, reco 49867; then the same cross-validated truncation
- Trade-off: correlations *between* the two tops stay at the physics order

Projection χ^2/ndf vs the $t\bar{t}$ fit sample (1/2)

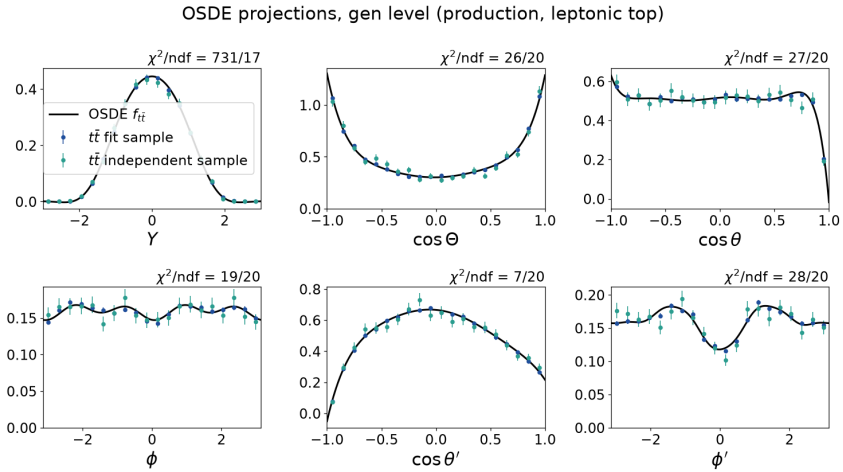
level	basis	Y	$\cos\Theta$	$\tilde{\phi}'$	$\cos\tilde{\theta}'$	$\tilde{\phi}$	$\cos\tilde{\theta}$	ϕ'	$\cos\theta'$	ϕ	$\cos\theta$
parton	paper	28.8	7.5	1.7	0.6	1.4	1.4	1.1	1.1	0.8	1.4
	paper, CV	28.6	6.9	1.9	0.6	1.4	1.6	1.3	1.1	0.9	1.4
	extended	0.5	0.5	1.5	0.4	1.1	1.2	0.9	0.8	0.6	1.2
	extended, CV	0.5	0.8	1.7	0.4	1.2	1.3	1.1	0.8	0.7	1.2
parton+trig	paper	7.9	2.7	1.3	0.3	1.1	1.1	47.0	2.5	0.8	2.7
	paper, CV	7.8	2.7	1.4	0.3	1.2	1.1	45.7	2.5	0.9	2.7
	extended	0.8	0.6	0.9	0.2	0.4	0.8	0.6	0.7	0.3	0.7
	extended, CV	0.8	1.0	1.4	0.3	1.2	1.1	1.8	1.4	0.4	0.9
gen	paper	1057.0	4.5	32.4	37.5	8.0	44.7	29.7	40.9	5.3	68.3
	paper, CV	906.9	4.6	32.5	39.4	8.2	44.7	30.1	40.9	5.6	68.3
	extended	16.1	1.3	0.4	0.5	0.2	0.2	0.2	0.4	0.1	0.7
	extended, CV	43.0	1.3	1.0	0.7	1.0	0.6	1.4	0.4	0.9	1.3

Projection χ^2/ndf vs the $t\bar{t}$ fit sample (2/2)

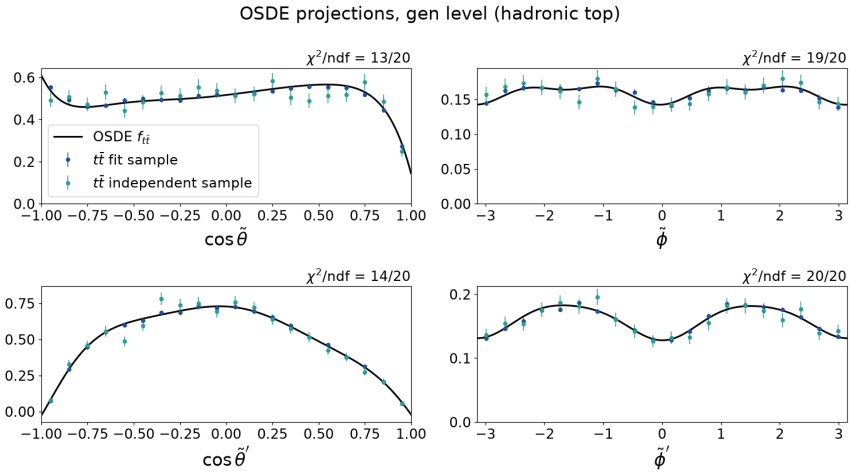
level	basis	Y	$\cos\Theta$	$\tilde{\phi}'$	$\cos\tilde{\theta}'$	$\tilde{\phi}$	$\cos\tilde{\theta}$	ϕ'	$\cos\theta'$	ϕ	$\cos\theta$
reco	paper	473.8	3.4	22.1	24.6	2.1	25.8	9.1	66.9	2.9	31.2
	paper, CV	473.3	3.6	22.1	24.2	2.2	26.6	9.1	68.8	2.9	31.2
	extended	9.6	0.4	0.3	0.7	0.2	1.1	0.5	0.4	0.1	0.6
	extended, CV	1.6	0.9	1.6	2.4	1.1	1.7	1.5	1.3	0.8	1.8

Extended basis (after CV truncation): χ^2/ndf $\approx$ 0.4–2 for every angle at every level; the paper basis is good only at parton level (apart from Y, $\cos\Theta$).

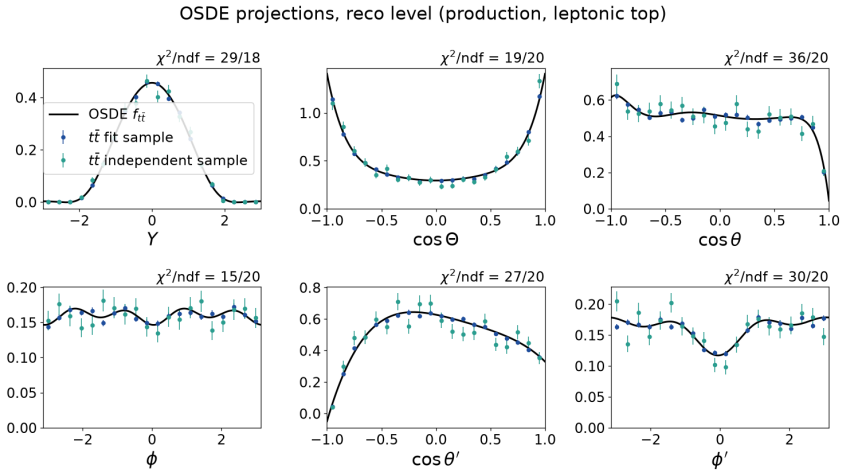
Extended $f_{t\bar{t}}^{\text{gen}}$ (CV-truncated): projections



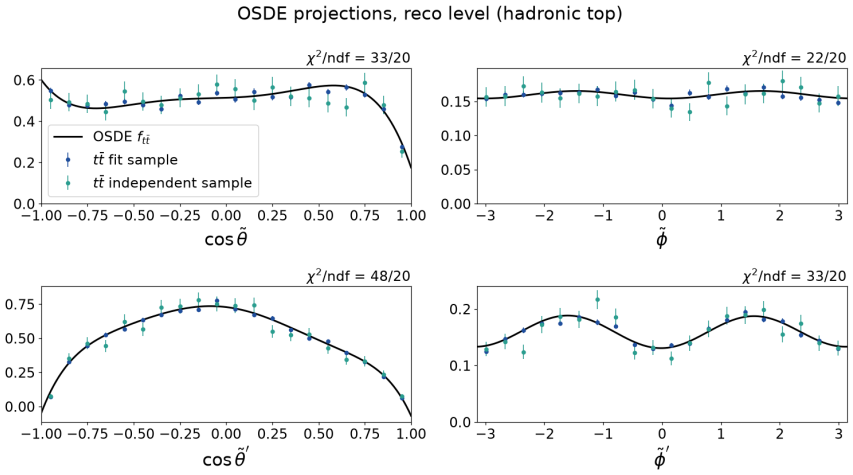
Extended $f_{t\bar{t}}^{\text{gen}}$ (CV-truncated): hadronic top



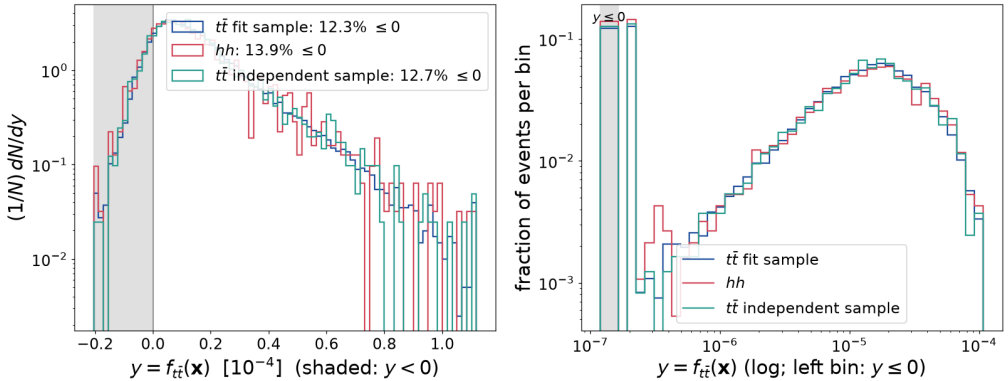
Extended $f_{t\bar{t}}^{\text{reco}}$ (CV-truncated): projections



Extended $f_{t\bar{t}}^{\text{reco}}$ (CV-truncated): hadronic top

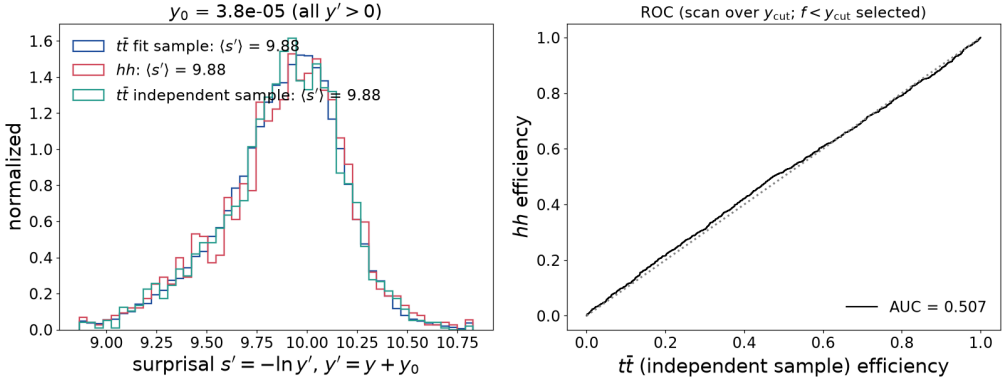


Extended $f_{t\bar{t}}^{\text{reco}}$ applied to reco-level $t\bar{t}$ and hh: y



Left: $y = f_{t\bar{t}}(\mathbf{x})$, linear (shaded: $y < 0$). Right: log scale; the left bin holds all events with $y \leq 0$. $t\bar{t}$: independent sample and fit sample (leave-one-out).

Extended $f_{t\bar{t}}^{\text{reco}}$ applied to reco-level $t\bar{t}$ and hh: surprisal and ROC



Left: $s' = -\ln(y + y_0)$, shifted so every event has $y' > 0$. Right: ROC from a scan over y_{cut} (events with $f < y_{\text{cut}}$ called hh-like).

Extended basis: performance

level	terms	number	AUC (hh vs $t\bar{t}$)	$f \leq 0$ ($t\bar{t}$ indep.)	$f \leq 0$ (hh)
parton	all terms	14436	0.655	1.4%	5.8%
	CV best	333	0.666	0.3%	2.5%
	5σ	57	0.669	0.1%	2.3%
parton+trig	all terms	18936	0.667	2.1%	8.2%
	CV best	498	0.684	0.4%	4.3%
	5σ	133	0.691	0.4%	4.3%
gen	all terms	32494	0.528	5.7%	8.4%
	CV best	679	0.535	1.6%	2.4%
	5σ	185	0.537	1.5%	2.2%
reco	all terms	49867	0.507	12.7%	13.9%
	CV best	323	0.512	1.4%	1.8%
	5σ	139	0.519	1.6%	1.7%

A much better description of $t\bar{t}$ hardly changes the hh separation at gen/reco: the limitation is the $t\bar{t}$-hypothesis reconstruction, not the density. Without truncation the many high-order terms add noise ($f \leq 0$ up to 13%).

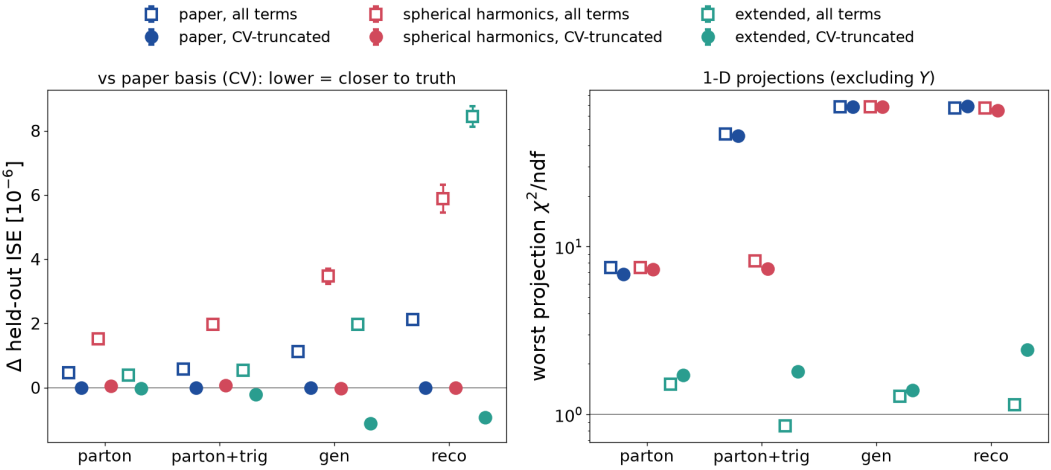
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2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Spherical-harmonic basis
9. Is the OSDE approach helping? (ML)
10. Why the separation washes out at gen level
11. Adding momentum to y: the resonance masses
12. Discriminator t (paper style)
13. Acceptance
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Spherical-harmonic products instead of the M functions

- Per top: $Y_{l_1}^{m_1}(\theta,\phi)\,Y_{l_2}^{m_2}(\theta',\phi')$, $l_1 \le 1$, $l_2 \le 2$: 36 functions (vs 24 M); with the same $p_l(\cos\Theta)$, $h_n(Y)$: 32 400 coefficients
- Orthonormal and closed under conjugation (tests); same OSDE, leave-one-out and CV truncation
- Physics difference: M uses spin-weighted $D^j_{m'm}(\phi,\theta,0)$; e.g. $d^1_{01} \propto \sin\theta$ is not a finite sum of Y_{l0}, so in principle the SM decay density, finite in M, needs infinitely many Y_{lm} products (tested below at 100k events)
- Figure of merit: held-out ISE $= \sum |\hat{c}|^2 - 2\langle \hat{f} \rangle_{\text{indep}}$ $(= \int (\hat{f} - f)^2 - \int f^2$; lower $=$ closer to the true density), on the independent $t\bar{t}$ sample

Paper (M) vs spherical harmonics vs extended



Basis comparison (independent $t\bar{t}$ sample) (1/2)

level	basis	terms	ΔISE vs M $[10^{-6}]$	worst χ^2/ndf	$f \leq 0$	AUC
parton	paper (M), all	14400	–	7.5	1.6%	0.653
	paper (M), CV	326	–	6.9	0.2%	0.665
	sph. harmonics, all	32400	$+1.05 \pm 0.07$	7.5	4.0%	0.639
	sph. harmonics, CV	150	$+0.06 \pm 0.02$	7.3	0.1%	0.670
	extended, all	14436	-0.08 ± 0.01	1.5	1.4%	0.655
	extended, CV	333	-0.02 ± 0.01	1.7	0.3%	0.666
parton+trig	paper (M), all	14400	–	47.0	1.6%	0.669
	paper (M), CV	385	–	45.7	0.4%	0.685
	sph. harmonics, all	32400	$+1.40 \pm 0.11$	8.2	4.5%	0.648
	sph. harmonics, CV	269	$+0.07 \pm 0.04$	7.4	0.3%	0.688
	extended, all	18936	-0.04 ± 0.04	0.9	2.1%	0.667
	extended, CV	498	-0.21 ± 0.03	1.8	0.4%	0.684

Basis comparison (independent $t\bar{t}$ sample) (2/2)

level	basis	terms	ΔISE vs M $[10^{-6}]$	worst χ^2/ndf	$f \leq 0$	AUC
gen	paper (M), all	14400	–	68.3	2.7%	0.525
	paper (M), CV	459	–	68.3	0.9%	0.525
	sph. harmonics, all	32400	$+2.35 \pm 0.23$	68.3	6.6%	0.520
	sph. harmonics, CV	555	-0.02 ± 0.11	68.3	1.3%	0.525
	extended, all	32494	$+0.85 \pm 0.15$	1.3	5.7%	0.528
	extended, CV	679	-1.11 ± 0.08	1.4	1.6%	0.535
reco	paper (M), all	14400	–	66.9	4.5%	0.507
	paper (M), CV	342	–	68.8	1.1%	0.505
	sph. harmonics, all	32400	$+3.77 \pm 0.43$	66.9	9.9%	0.507
	sph. harmonics, CV	279	-0.01 ± 0.14	64.5	1.5%	0.510
	extended, all	49867	$+6.34 \pm 0.32$	1.1	12.7%	0.507
	extended, CV	323	-0.92 ± 0.11	2.4	1.4%	0.512

ΔISE < 0: closer to the true density than the paper's M basis (paired, same events). Worst χ^2/ndf of the 1-D projections excludes Y.

Basis comparison: conclusions

- All terms kept: spherical harmonics are worse than M at every level (9–16 σ in ΔISE): 32 400 terms, most of them noise
- After CV truncation: parton ΔISE(SH) $= +0.06 \pm 0.02$; parton+trig ΔISE(SH) $= +0.07 \pm 0.04$; gen ΔISE(SH) $= -0.02 \pm 0.11$; reco ΔISE(SH) $= -0.01 \pm 0.14$ (10^{-6}): spherical harmonics and M describe $t\bar{t}$ about equally well
- Spherical harmonics fit the trigger-sculpted ϕ' better (parton+trig worst χ^2/ndf 7.4 vs 45.7): $l_2 = 2$ allows a second harmonic in ϕ', which M ties to $j_1 \leq 1$
- The extended basis is best at gen/reco: parton ΔISE $= -0.02 \pm 0.01$; parton+trig ΔISE $= -0.21 \pm 0.03$; gen ΔISE $= -1.11 \pm 0.08$; reco ΔISE $= -0.92 \pm 0.11$
- The hh AUC hardly depends on the basis

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12. Discriminator t (paper style)
13. Acceptance
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Setup: same events, same inputs

- Test set for every method: the independent $t\bar{t}$ sample (10000) and a held-out 40% of hh (8000); training: $t\bar{t}$ fit sample + 50% of hh (10% each for validation)
- **Unsupervised** (trained on $t\bar{t}$ only, like $f_{t\bar{t}}$): Gaussian-mixture density, autoencoder, normalizing flow (RealNVP) on the same ten angles
- **Supervised** ($t\bar{t}$ vs hh labels; an upper bound): BDT, MLP, FT-Transformer on (a) our angles, (b) angles + parton momenta; particle transformer (pairwise m, ΔR) on momenta
- **Does y help ML?** the same classifiers with $y = f_{t\bar{t}}$ and s' added (leave-one-out y for $t\bar{t}$ training events)
- Controls: shuffled labels and $t\bar{t}$-vs-$t\bar{t}$ must give AUC ≈ 0.5; train vs test AUC

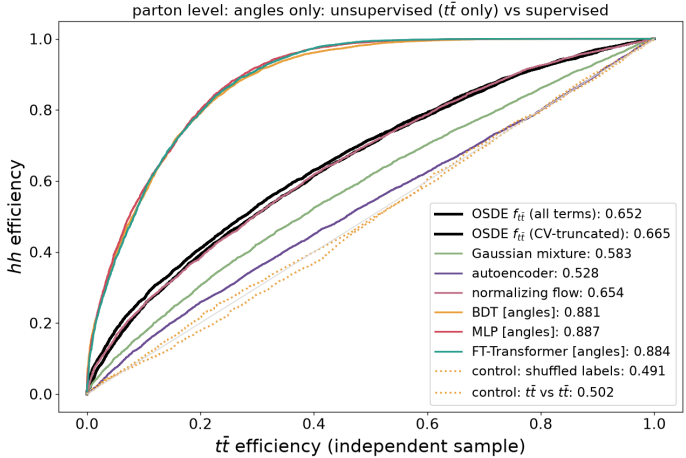
The ML methods in plain words: unsupervised (see only $t\bar{t}$)

- **Same job as** $f_{t\bar{t}}$: learn the $t\bar{t}$ density from $t\bar{t}$ events; low density $\Rightarrow$ not $t\bar{t}$
- **Gaussian mixture**: $f_{t\bar{t}}$ = a sum of 32 Gaussian blobs in the 10-D angle space; blob positions and widths fitted to $t\bar{t}$
- **Autoencoder**: a network squeezes the 10 angles through a few numbers and rebuilds them; trained on $t\bar{t}$, so events it rebuilds badly are unusual. Not a density
- **Normalizing flow**: learn a smooth, invertible map $z = T(x)$ that turns the $t\bar{t}$ distribution into a plain Gaussian. Then $f_{t\bar{t}}(x) = \mathcal{N}(T(x))\,|\det \partial T/\partial x|$
- The mass map used later ($u = \frac{1}{2} + \frac{1}{\pi}\arctan\frac{m-M}{W}$, density $g_u\,du/dm$) is exactly a 1-D flow; the normalizing flow learns the same thing in 10-D. RealNVP stacks 8 simple steps, each shifting and rescaling half of the variables by amounts computed from the other half, so the Jacobian stays easy

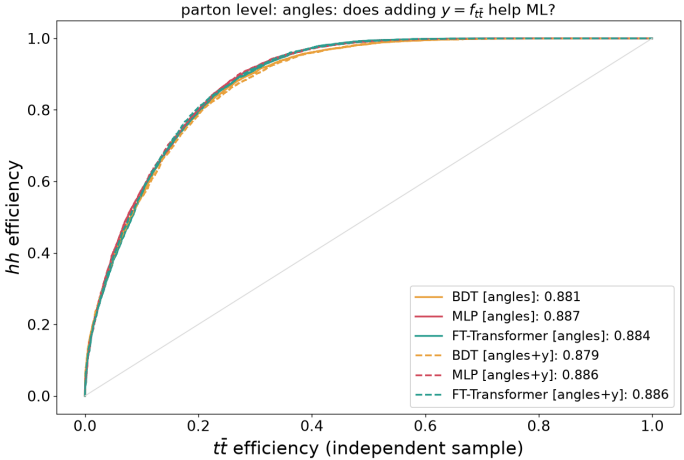
The ML methods in plain words: supervised (see $t\bar{t}$ and hh)

- **Different job**: learn $P(hh \,|\, x)$ from labelled examples. The best possible is the ratio $f_{hh}/f_{t\bar{t}}$; $f_{t\bar{t}}$ alone assumes nothing about hh (as if f_{hh} were flat)
- **BDT**: a few hundred small decision trees ("if $\cos\theta < -0.6$ and $|Y| > 1\ldots$"), each fixing the mistakes of the previous ones
- **MLP**: a neural network; layers of weighted sums with a simple non-linearity in between
- **FT-Transformer**: each input variable becomes a vector ("token"); attention lets every variable look at every other one, so it learns which combinations matter
- **Particle Transformer**: the tokens are the six particles' four-momenta; attention is told each pair's invariant mass and ΔR, so it sees $m_{b\bar{b}} = m_h$ directly
- Why they win: they know what hh looks like. Our goal is different: a $t\bar{t}$-only (unsupervised) y that flags anything non-$t\bar{t}$

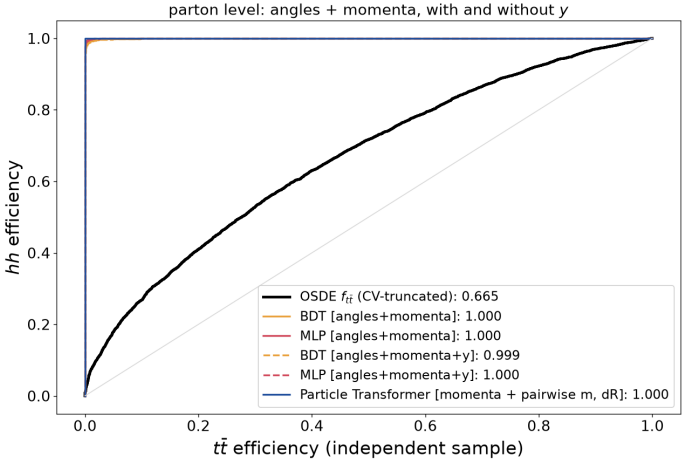
ROC, parton level: unsupervised vs supervised, angles



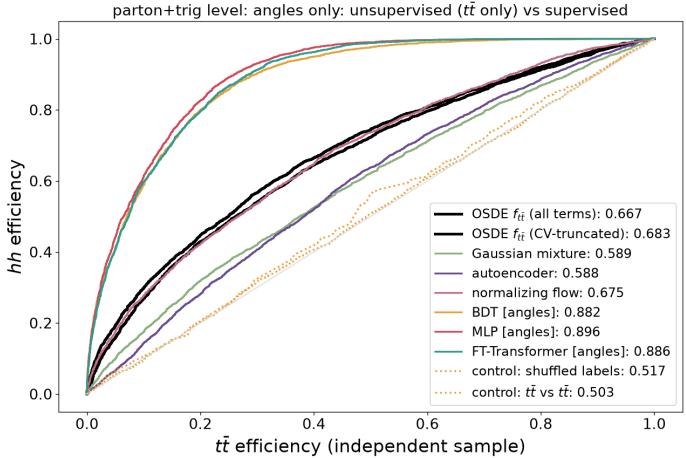
ROC, parton level: does y help?



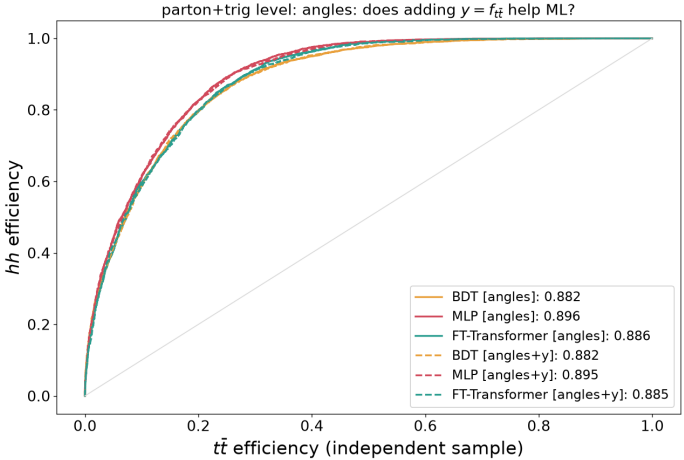
ROC, parton level: angles + momenta



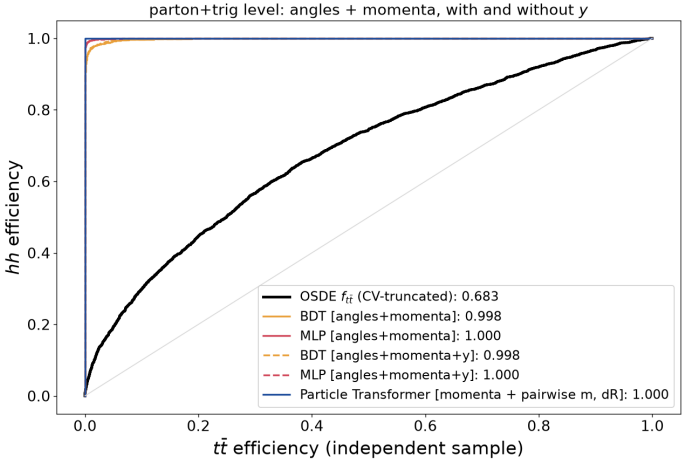
ROC, parton+trig level: unsupervised vs supervised, angles



ROC, parton+trig level: does y help?



ROC, parton+trig level: angles + momenta



AUC (hh vs $t\bar{t}$), parton level (1/2)

inputs	method	AUC (test)	AUC (train)
unsupervised	OSDE $f_{t\bar{t}}$, paper basis (all terms)	0.652 ± 0.004	–
unsupervised	OSDE $f_{t\bar{t}}$, paper basis (CV-truncated)	0.665 ± 0.004	–
unsupervised	Gaussian mixture density (32 comp.)	0.583 ± 0.005	–
unsupervised	autoencoder (reconstruction error)	0.528 ± 0.004	–
unsupervised	normalizing flow (RealNVP, 8 couplings)	0.654 ± 0.004	–
angles	BDT	0.881 ± 0.002	0.921
angles	MLP	0.887 ± 0.002	0.905
angles	FT-Transformer	0.884 ± 0.002	0.894
angles + y	BDT	0.879 ± 0.002	0.918
angles + y	MLP	0.886 ± 0.002	0.906
angles + y	FT-Transformer	0.886 ± 0.002	0.894
angles + momenta	BDT	1.000 ± 0.000	1.000

AUC (hh vs $t\bar{t}$), parton level (2/2)

inputs	method	AUC (test)	AUC (train)
angles + momenta	MLP	1.000 ± 0.000	1.000
angles + momenta + y	BDT	0.999 ± 0.000	1.000
angles + momenta + y	MLP	1.000 ± 0.000	1.000
momenta	Particle Transformer	1.000 ± 0.000	1.000
control	control: BDT, shuffled labels	0.491 ± 0.005	–
control	control: BDT, tt (fit) vs tt (independent)	0.502 ± 0.006	–

Answer, parton level

- As an anomaly detector (no hh used): OSDE $f_{t\bar{t}}$ AUC 0.665 vs 0.654 for the best other unsupervised method (normalizing flow); paired difference $+0.011 \pm 0.003$
- Knowing hh (supervised) on the same angles: AUC 0.887, the upper bound for these inputs
- Adding y to the supervised classifiers: AUC change -0.002 to $+0.002$ (best 0.886); test-AUC error ± 0.002: y carries no information the classifier does not already learn from the angles
- Momenta (which contain $m_{b\bar{b}} = m_h$ etc.) make the task trivial: AUC 0.9999
- Controls: shuffled labels 0.491, $t\bar{t}$ vs $t\bar{t}$ 0.502 (expected 0.5)

AUC (hh vs $t\bar{t}$), parton+trig level (1/2)

inputs	method	AUC (test)	AUC (train)
unsupervised	OSDE $f_{t\bar{t}}$, paper basis (all terms)	0.667 ± 0.005	–
unsupervised	OSDE $f_{t\bar{t}}$, paper basis (CV-truncated)	0.683 ± 0.005	–
unsupervised	Gaussian mixture density (32 comp.)	0.589 ± 0.006	–
unsupervised	autoencoder (reconstruction error)	0.588 ± 0.006	–
unsupervised	normalizing flow (RealNVP, 8 couplings)	0.675 ± 0.005	–
angles	BDT	0.882 ± 0.003	0.936
angles	MLP	0.896 ± 0.003	0.916
angles	FT-Transformer	0.886 ± 0.003	0.901
angles + y	BDT	0.882 ± 0.003	0.938
angles + y	MLP	0.895 ± 0.003	0.912
angles + y	FT-Transformer	0.885 ± 0.003	0.898
angles + momenta	BDT	0.998 ± 0.000	1.000

AUC (hh vs $t\bar{t}$), parton+trig level (2/2)

inputs	method	AUC (test)	AUC (train)
angles + momenta	MLP	1.000 ± 0.000	1.000
angles + momenta + y	BDT	0.998 ± 0.000	1.000
angles + momenta + y	MLP	1.000 ± 0.000	1.000
momenta	Particle Transformer	1.000 ± 0.000	1.000
control	control: BDT, shuffled labels	0.517 ± 0.005	–
control	control: BDT, tt (fit) vs tt (independent)	0.503 ± 0.007	–

Answer, parton+trig level

- As an anomaly detector (no hh used): OSDE $f_{t\bar{t}}$ AUC 0.683 vs 0.675 for the best other unsupervised method (normalizing flow); paired difference $+0.008 \pm 0.003$
- Knowing hh (supervised) on the same angles: AUC 0.896, the upper bound for these inputs
- Adding y to the supervised classifiers: AUC change -0.001 to -0.000 (best 0.895); test-AUC error ± 0.003: y carries no information the classifier does not already learn from the angles
- Momenta (which contain $m_{b\bar{b}} = m_h$ etc.) make the task trivial: AUC 0.9998
- Controls: shuffled labels 0.517, $t\bar{t}$ vs $t\bar{t}$ 0.503 (expected 0.5)

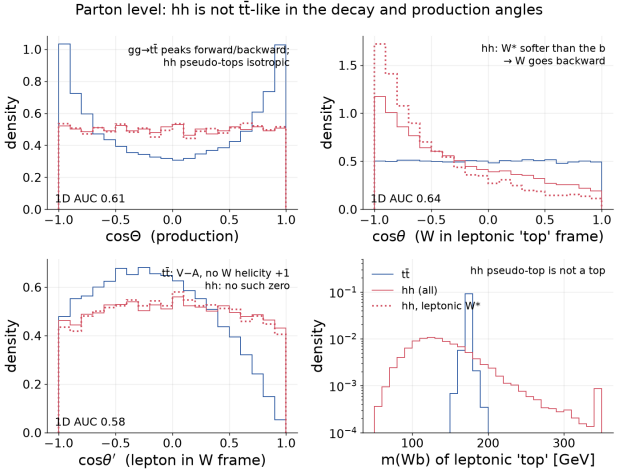
Outline

1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Spherical-harmonic basis
9. Is the OSDE approach helping? (ML)
10. Why the separation washes out at gen level
11. Adding momentum to y: the resonance masses
12. Discriminator t (paper style)
13. Acceptance
14. Study A′: hh with on-shell W bosons ($m_H = 170$ GeV)
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16. Pileup
17. Level-1 trigger: a lookup-table version of angles + BW
18. Summary

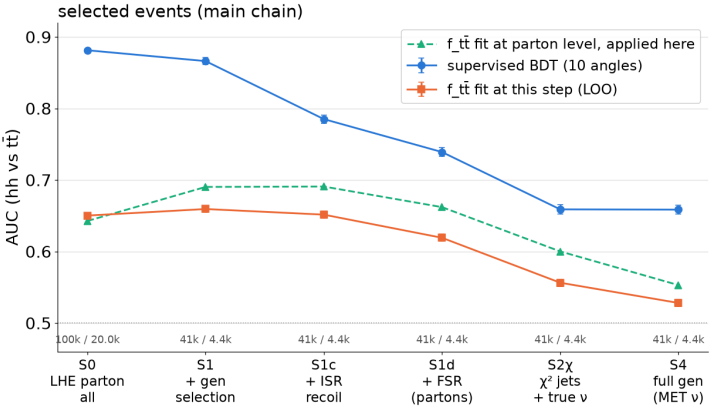
Short answer

- At parton level the signal "tops" are not tops: each is the W from one Higgs plus the b from the other (m_{Wb} median 136 GeV), and one W is off-shell (m_{W^*} median 32 GeV)
- $\Rightarrow$ the signal has $\cos\theta$, $\cos\tilde{\theta}$ peaked at -1, isotropic $\cos\Theta$ (no t-channel), and no $V-A$ zero in $\cos\theta'$
- At gen level the $t\bar{t}$-hypothesis reconstruction rebuilds m_W and m_t from whatever jets fit: the χ^2 picks the true jets in 8% of the signal (34% of $t\bar{t}$); the m_W-constrained ν puts the leptonic W^* on shell
- Ranked losses in $f_{t\bar{t}}$ AUC: wrong jets in the χ^2 (-0.06), FSR (-0.03), ν from m_W (-0.03); ISR recoil and selection cost nothing
- The information is **not** gone: a BDT on hypothesis-free gen observables (m_{bb}, ΔR_{bb}, MET, m_T, ...) gives AUC 0.87 on the same events

Parton level: what separates hh from $t\bar{t}$

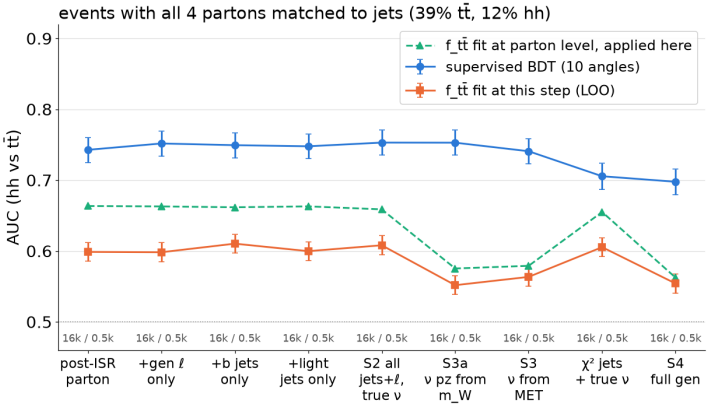


Step by step from parton to gen: all selected events



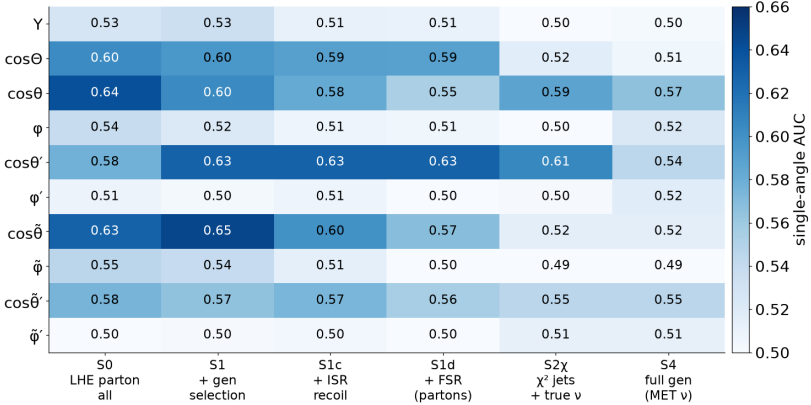
Green: parton-level $f_{t\bar{t}}$ applied unchanged; orange: $f_{t\bar{t}}$ refit at each step; blue: supervised BDT on the ten angles. Biggest $f_{t\bar{t}}$ loss: the χ^2 jet choice.

Step by step: events with correctly matched jets



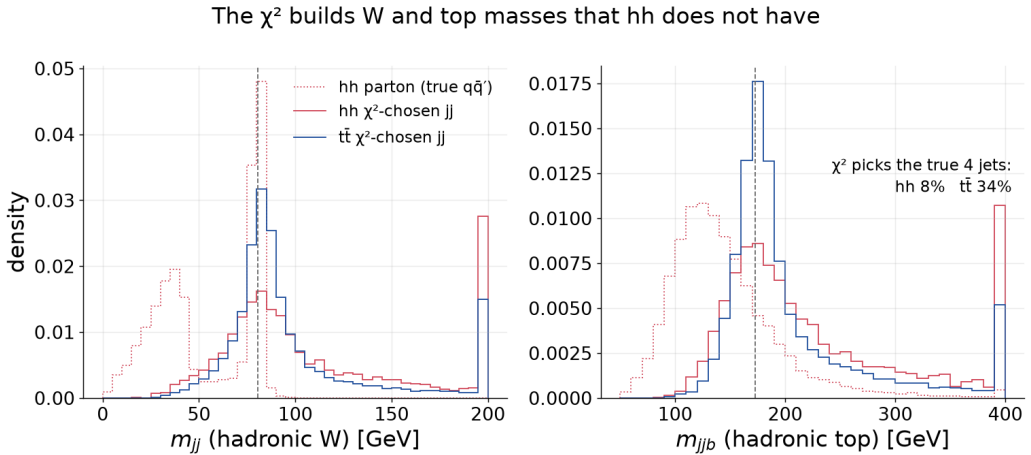
When all four partons match distinct jets, jets cost nothing; the m_W-constrained ν (choice of p_z solution) costs 0.05–0.08.

Which angles lose the separation, and where

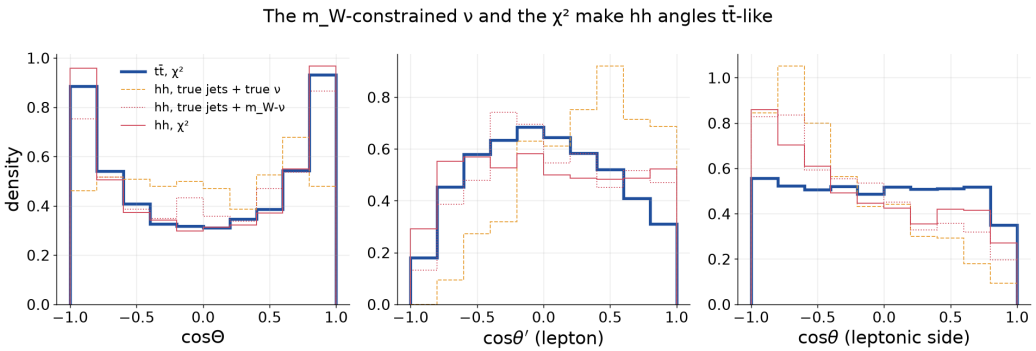


$\cos\Theta$ and $\cos\tilde{\theta}$ die at the χ^2 jet choice; $\cos\theta'$ dies with the m_W-constrained neutrino; ISR recoil mainly costs ϕ, $\tilde{\phi}$, Y.

The χ^2 builds W and top masses that hh does not have

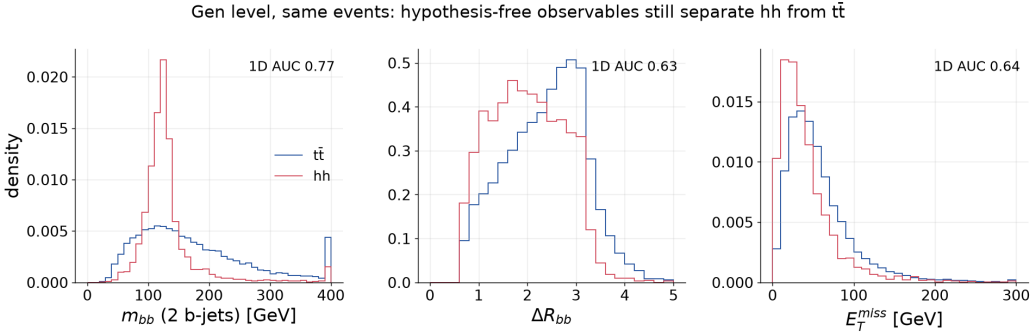


hh angles become $t\bar{t}$-like

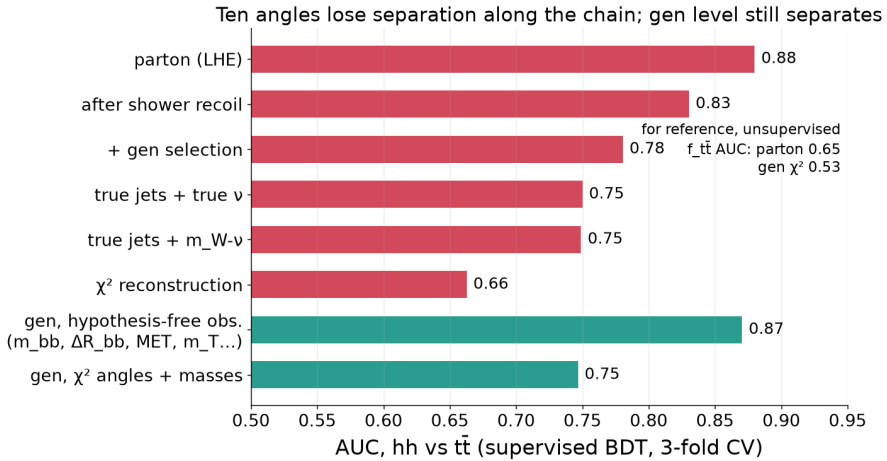


"true jets + true ν" is a small, biased subset (532 hh events).

Gen level still separates, with hypothesis-free observables



Summary of the chain (supervised BDT)



Implications

- The angles are the right variables for $t\bar{t}$, but at gen/reco they are computed under the $t\bar{t}$ hypothesis, which removes exactly what makes hh different
- $f_{t\bar{t}}$ is unsupervised: it flags only where $t\bar{t}$ is rare; the fitted reconstruction moves hh into the $t\bar{t}$-dense region
- Options: (i) add the χ^2 value and the masses (m_{bb}, m_{jj}, m_{jjb}) as extra dimensions of $f_{t\bar{t}}$; (ii) use hh-hypothesis angles; (iii) loosen the m_W constraint for the ν
- Caveats: matched hh subsets are small (500 events); FSR step uses last-copy partons (slight overestimate)

Outline

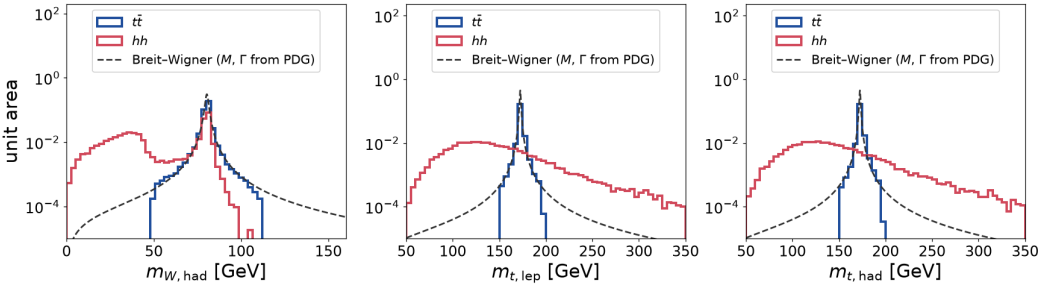
1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
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Motivation: what the ten angles leave out

- 6 final-state particles = 18 momentum components = 10 paper variables (Y, Θ, 8 decay angles) + 4 **resonance masses** ($m_{W,\ell}, m_{W,h}, m_{t,\ell}, m_{t,h}$) + $m_{t\bar{t}}$, $\vec{p}_T^{\,t\bar{t}}$, overall azimuth
- For $t\bar{t}$ the masses carry no information: each is a Breit–Wigner fixed by M and Γ, and (narrow width) $|\mathcal{M}|^2$ factorizes: angles $\times \prod_k \mathrm{BW}(m_k)$. So the paper could drop them
- hh is *not* on that shell: one W is off shell (median 32 GeV) and its "tops" sit at 135 GeV. Its difference lives in exactly the dimensions we dropped

The dropped dimensions, parton level

parton level: $t\bar{t}$ sits on the mass shell (Breit–Wigner), hh does not

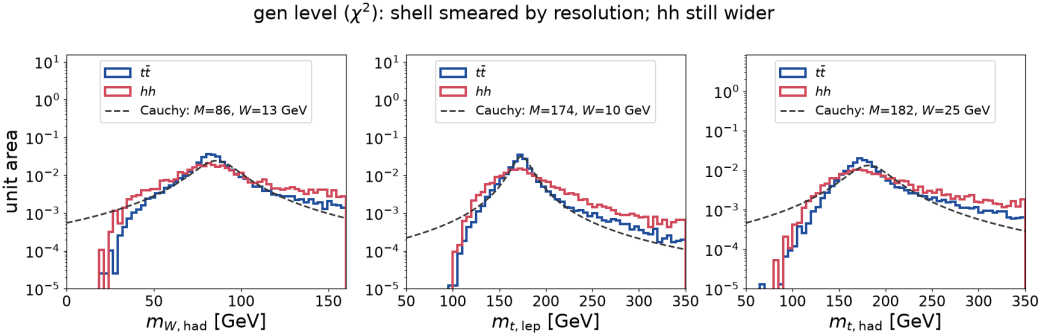


$t\bar{t}$ (blue) is the Breit–Wigner (dashed; MadGraph cuts it at $\pm 15\Gamma$). hh (red) is spread over 100 GeV.

The extension: one factor per mass, no new coefficients

- $y_{\text{ext}}(\Omega, m) = y(\Omega) \times \prod_{k=1}^{4} \mathrm{BW}_k(m_k)$ (the $t\bar{t}$ density on the full phase space)
- In surprisal (what t is built from) the factors simply add: $s_{\text{ext}} = s(\Omega) + \sum_k \ln\left[(m_k^2 - M_k^2)^2 + M_k^2\Gamma_k^2\right] + \text{const}$
- Parton level: M, Γ from the PDG. Nothing is fitted
- Gen/reco level: resolution replaces Γ; use the same shape with M, W = median and half interquartile range of $t\bar{t}$: two numbers per mass
- A fitted OSDE correction to this shape (Legendre in $u = \frac{1}{2} + \frac{1}{\pi}\arctan\frac{m-M}{W}$) changes the gen AUC by only $+0.004$

The same dimensions at gen level (χ^2 reconstruction)



Resolution widens the $t\bar{t}$ shell (dashed: two-number model); hh is still wider, mainly in the tails. $m_{W,\ell}$ is missing: the ν solution sets it to m_W by construction.

Result: AUC (hh vs $t\bar{t}$), unsupervised

	parton	gen (χ^2)
angles only, $y(\Omega)$ (paper)	0.648	0.528
$+ m_{W,\ell}$	0.786	(fixed by ν)
$+ m_{W,h}$	0.993	0.631
$+ m_{t,\ell}$	0.999	0.683
$+ m_{t,h}$ (all masses)	1.000	0.689

Gen level with the masses (0.69) beats the parton-level angles (0.65): the parton-level performance is regained. Rows add one mass at a time. Test: half of 40k $t\bar{t}$ (not used to set M, W; angles scored leave-one-out), 20k / 4.4k hh.

With the masses in, how many angular terms are needed?

angular terms kept	parton: terms	angles	+BW	gen: terms	angles	+BW
none (flat in angles)	0	–	1.000	0	–	0.689
$\lvert c\rvert/\sigma_c > 40$	7	0.653	1.000	5	0.546	0.692
$\lvert c\rvert/\sigma_c > 20$	16	0.662	1.000	14	0.555	0.692
$\lvert c\rvert/\sigma_c > 10$	23	0.665	1.000	54	0.534	0.689
$\lvert c\rvert/\sigma_c > 5$	54	0.663	1.000	153	0.528	0.688
$\lvert c\rvert/\sigma_c > 3$	186	0.661	1.000	330	0.528	0.688
CV (paper method)	326	0.661	1.000	459	0.529	0.688
all	14400	0.648	1.000	14400	0.528	0.689

Terms ranked by significance $|c|/\sigma_c$. For hh the masses do all the work: the angular part can be flat. Even for the angles alone, $\sim$20 terms beat all 14,400 (fewer noisy terms).

The $\sim$20 terms that matter (parton $t\bar{t}$)

- Production: $P_2(\cos\Theta)$, $P_4(\cos\Theta)$ (forward-backward peaking) and $H_2(Y)$, $H_4(Y)$ (rapidity width)
- Decay: $j_1 = 0$, $j_2 = 1, 2$ on each side, i.e. functions of $\cos\theta'$ only: the W helicity fractions (F_0, F_L, F_R)
- Their products (e.g. W helicity $\times P_2(\cos\Theta)$, lep $\times$ had): mostly what independent factors produce
- No top-polarization or spin-correlation term is in the top 23: LO QCD $t\bar{t}$ is unpolarized and its spin correlations are weak at this sample size
- Recommendation: $y = [\text{few-term angular } f] \times \prod_k \mathrm{BW}(m_k)$. Keep the angular part (cheap) because it, not the masses, is what sees an *on-shell* signal with different spin structure

Why do fewer terms do better? Estimation noise

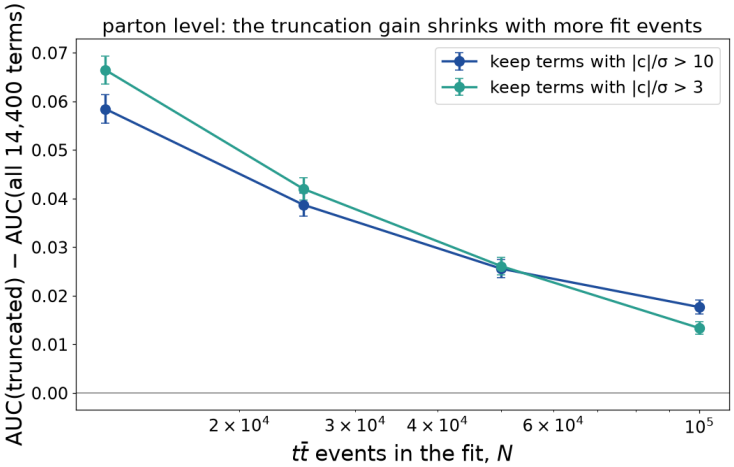
- Each $\hat{c}_k$ is a sample mean over N events: noise $\sigma_k \propto 1/\sqrt{N}$. Most of the 14,400 true coefficients are ≈ 0, so those terms contribute only noise
- Summed over thousands of terms, the noise is a random wobble on $\hat{f}(x)$, unrelated to $t\bar{t}$ vs hh: it scrambles the event ranking and lowers the AUC
- Dropping insignificant terms removes the wobble at the cost of a small bias: the usual bias–variance trade-off (what the CV truncation optimizes for the density)
- Test: refit on $N = 12.5$k–100k $t\bar{t}$ events (disjoint subsets), score the independent $t\bar{t}$ sample; the gain must shrink as N grows, and two independent fits must agree only for the truncated f

It is noise, not a fluctuation

fit events N	AUC, all terms	AUC, $\lvert c\rvert/\sigma > 10$	gain	score corr. of 2 disjoint fits: all / trunc.
12.5k	0.602	0.661 (10 terms)	$+0.058$	0.46 / 0.989
25k	0.627	0.666 (16 terms)	$+0.039$	0.65 / 0.998
50k	0.643	0.669 (19 terms)	$+0.026$	0.79 / 1.000
100k	0.653	0.670 (23 terms)	$+0.018$	–

Gain errors ± 0.002–0.003 (paired bootstrap). The truncated f is converged already at 12.5k; the full f approaches it as $1/\sqrt{N}$. Two disjoint full fits rank events differently (corr. 0.46 at 12.5k): most of the full expansion is noise.

Truncation gain vs fit-sample size



Momentum extension at gen and reco

level	$t\bar{t}$ / hh	angles (all)	angles ($> 10\sigma$)	masses (BW)	angles ($> 10\sigma$) + BW
gen	4134 / 4416	0.525	0.536	0.690	0.692
reco	2440 / 1878	0.507	0.518	0.600	0.602

Independent $t\bar{t}$ sample (no fit has seen it); M, W from 30k fit-sample events. Widths W gen/reco: $m_{W,h}$ 13/12, $m_{t,\ell}$ 10/16, $m_{t,h}$ 25/25 GeV. Reco loses to jet resolution and wrong-jet combinatorics; still $0.51 \to 0.60$.

Realistic quark tagging: what the paper assumed and what we do

- The hadronic primed angles need the *down-type* W daughter. The paper assumes it is known; so far we matched jets to the LHE quarks (perfect up/down tagging)
- Reality: b-tagging is already realistic at reco (Delphes ATLAS parametrization). The only up/down handle is c-tagging: in $W \to cs$ (50%) a c-tagged jet is up-type, so the other is the down-type s
- Instead of a hard choice: $w = P(\text{softer jet is down} \,|\, \text{c-tags, kinematics})$ by Bayes ($W \to cs$ or ud, tag efficiencies, prior $P(\text{softer is down}) = 58\%$ parton, 53% reco)
- $y = w\, f(\Omega_{\text{down=softer}}) + (1-w)\, f(\Omega_{\text{down=harder}})$: the $t\bar{t}$ density of the event as observed. f is unchanged (fit with truth labels)
- Tags emulated from the true flavour (Delphes jet flavour; quark id at parton). Working points illustrative, ATLAS-like

Effect of realistic c-tagging (AUC hh vs $t\bar{t}$)

down-type assignment	label right ($t\bar{t}$)	parton: angles	+BW	reco: angles	+BW
perfect up/down tag (paper)	100%	0.670	1.000	0.518	0.602
ideal c-tag	79%	0.661	1.000	0.517	0.602
c-tag loose (45/30/8%)	65%	0.653	1.000	0.517	0.602
c-tag tight (30/11/1.4%)	64%	0.652	1.000	0.517	0.602
no tag: average both labels	–	0.650	1.000	0.517	0.602
no tag: softer jet = down	58%	0.643	1.000	0.515	0.601

Angular f with $|c|/\sigma > 10$ terms; tagging working points: ϵ_c / b / light mistag. Without the label the angles lose at most 0.02 at parton and nothing at reco; with the masses nothing at all (the masses do not depend on which jet is down-type).

Momentum in y: summary and next steps

- Physics: hh differs from $t\bar{t}$ mostly *off* the $t\bar{t}$ mass shell; the angles alone cannot see it
- Simplest fix: multiply y by one Breit–Wigner per resonance (2 numbers each at gen). Nothing else changes; t is built from the summed surprisal as before
- Parton: 0.65 $\rightarrow$ 1.00; gen: 0.53 $\rightarrow$ 0.69
- Reco: 0.51 $\rightarrow$ 0.60; realistic c-tagging changes nothing once the masses are in
- What the angles add on top of BW: in the product model nothing ($+0.002$ gen, $+0.002$ reco); a supervised BDT on angles+masses vs masses gains $+0.029 \pm 0.004$ (gen), $+0.037 \pm 0.015$ (reco): a little is there for a joint fit
- Next: solve the ν with the $m_{t,\ell}$ constraint (frees $m_{W,\ell}$); add $m_{t\bar{t}}$; pileup

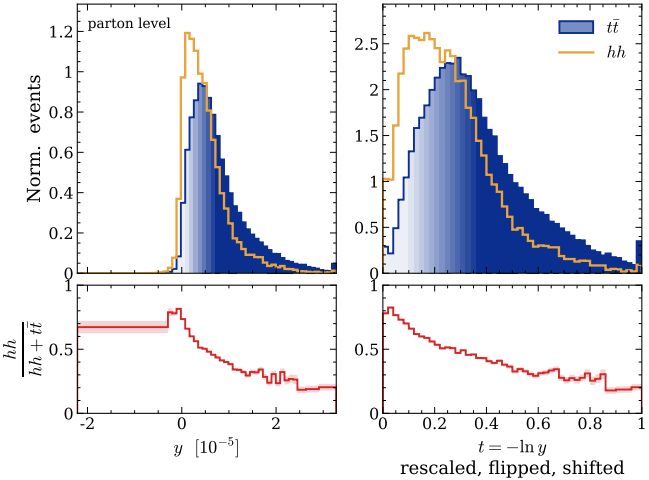
Outline

1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Spherical-harmonic basis
9. Is the OSDE approach helping? (ML)
10. Why the separation washes out at gen level
11. Adding momentum to y: the resonance masses
12. Discriminator t (paper style)
13. Acceptance
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16. Pileup
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18. Summary

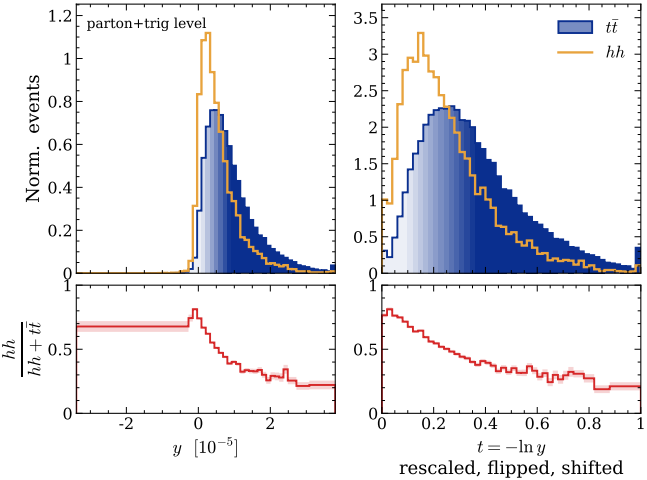
Definition of t (as in the paper)

- $y' = y - y_0$, $s' = -\ln y'$, $t = (s'_1 - s')/(s'_1 - s'_0) \in [0, 1]$; $t \to 0$: less $t\bar{t}$-like
- $y_0 = 2.5 \times \min y$: the paper figure's axis starts at $\approx 2.5 \times \min y$ and its t is populated from 0, which needs y_0 well below $\min y$
- s'_0, s'_1 = 0.5%, 99.5% quantiles of s' ($t\bar{t} + hh$); events beyond go into the first/last bin, so t fills [0, 1] with no empty bins
- $t\bar{t}$: fit sample with leave-one-out y; both samples at unit area; ratio $hh/(hh + t\bar{t})$ below

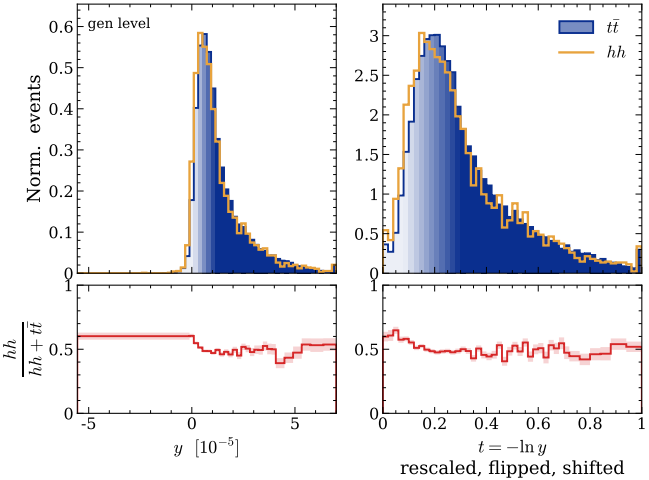
$y = f_{t\bar{t}}^{\text{parton}}$ and t, parton level (paper basis)



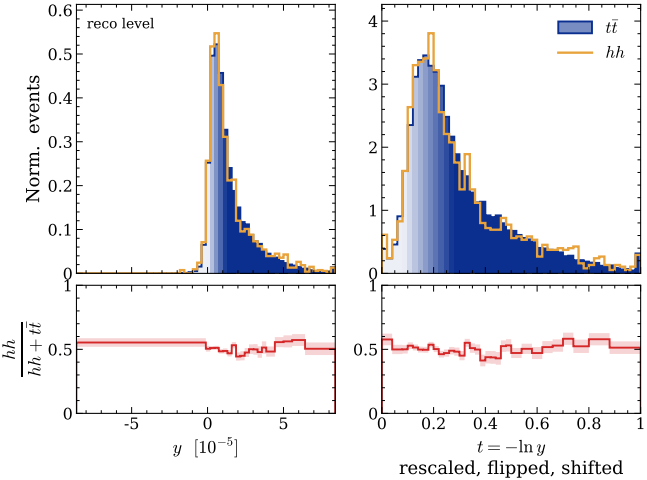
$y = f_{t\bar{t}}^{\text{parton+trig}}$ and t, parton+trig level (paper basis)



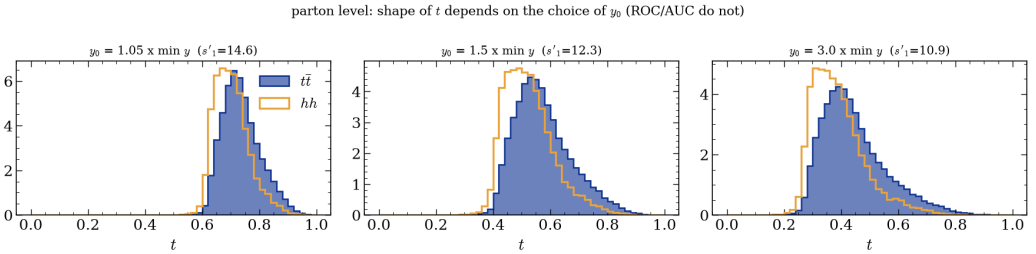
$y = f_{t\bar{t}}^{\text{gen}}$ and t, gen level (paper basis)



$y = f_{t\bar{t}}^{\text{reco}}$ and t, reco level (paper basis)



Shape of t depends on y_0



s'_1 is set by the most negative event, so y_0 just below $\min y$ squeezes t toward 1. The paper figure (axis from -1.5×10^{-5}, t populated from 0) implies $y_0 \approx 2.5 \times \min y$, used here. The ROC/AUC do not depend on y_0 (monotonic).

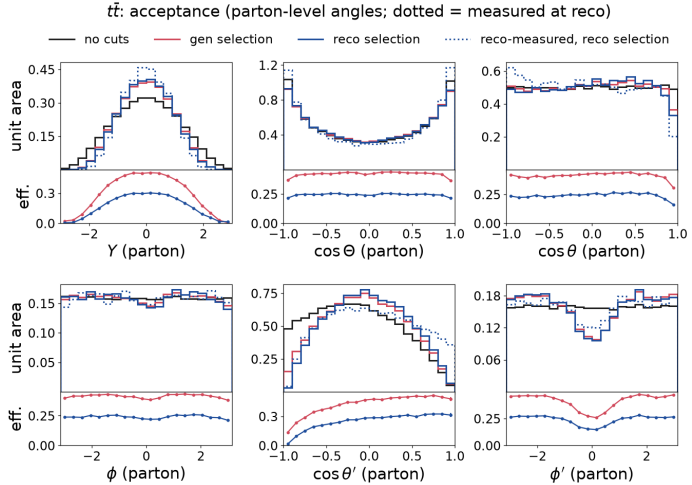
Outline

1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Spherical-harmonic basis
9. Is the OSDE approach helping? (ML)
10. Why the separation washes out at gen level
11. Adding momentum to y: the resonance masses
12. Discriminator t (paper style)
13. Acceptance
14. Study A′: hh with on-shell W bosons ($m_H = 170$ GeV)
15. Study B: $t\bar{t}$ vs $H \to t\bar{t}$
16. Pileup
17. Level-1 trigger: a lookup-table version of angles + BW
18. Summary

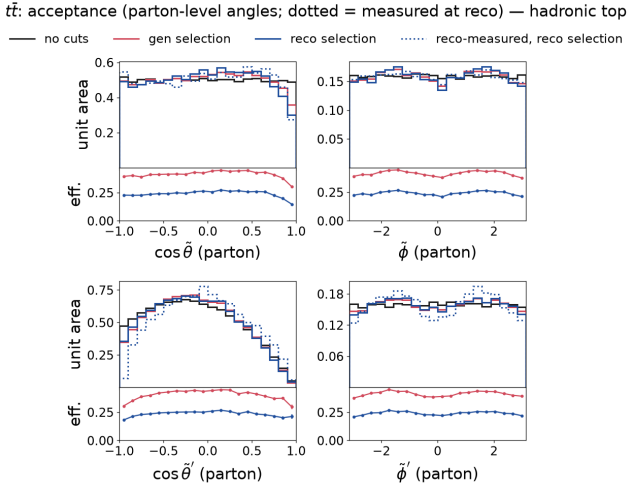
Acceptance: how the selection sculpts the angles

- Same (parton-level) angle values for every curve, so shape changes are acceptance only; dotted = angles measured at reco level (adds resolution and jet assignment)
- $\cos\theta' \to -1$, $\phi' \approx 0$: the lepton is emitted against the top's flight direction and fails $p_T > 25$ GeV (already at gen level, so not isolation)
- $|Y| \gtrsim 2$: $|\eta| < 2.5$ acceptance. $\cos\theta, \cos\tilde{\theta} \to +1$: the b quark goes backward and is too soft for the 25 GeV jet cut
- $\cos\Theta$ and the ϕ angles: nearly flat efficiency
- hh: the $t\bar{t}$-hypothesis χ^2 turns flat parton $\cos\Theta$ into $t\bar{t}$-like peaks at ± 1 (dotted): the reconstruction, more than the acceptance, makes hh look like $t\bar{t}$

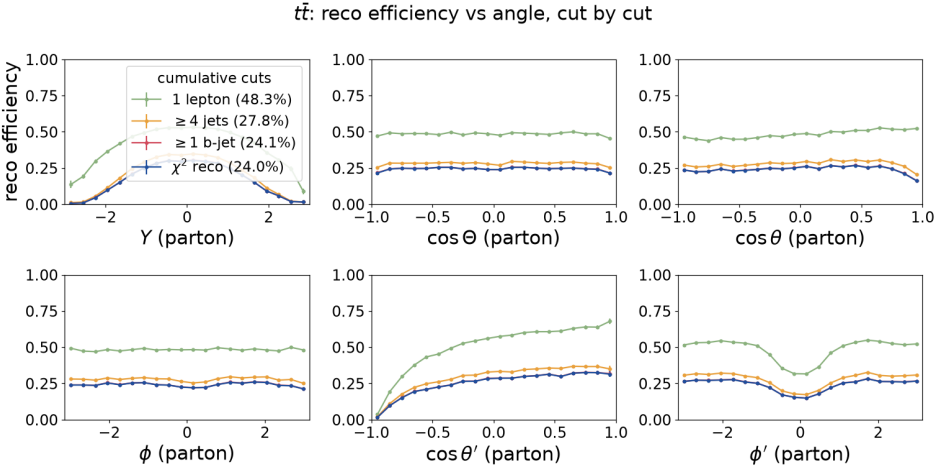
Acceptance, $t\bar{t}$: no cuts vs after cuts (production, leptonic top)



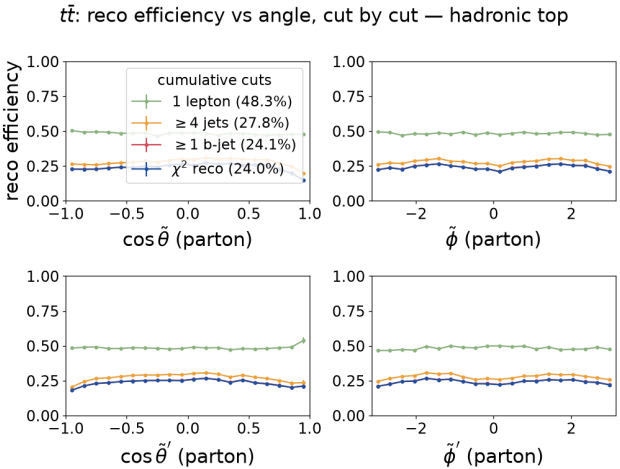
Acceptance, $t\bar{t}$: no cuts vs after cuts (hadronic top)



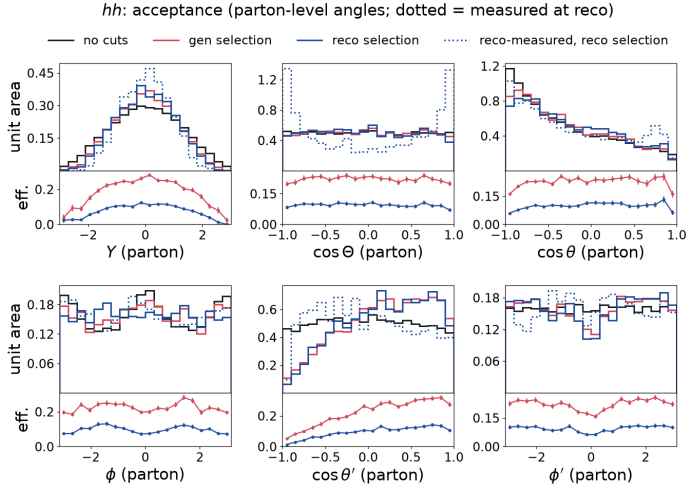
Acceptance, $t\bar{t}$: efficiency vs angle, cut by cut (production, leptonic top)



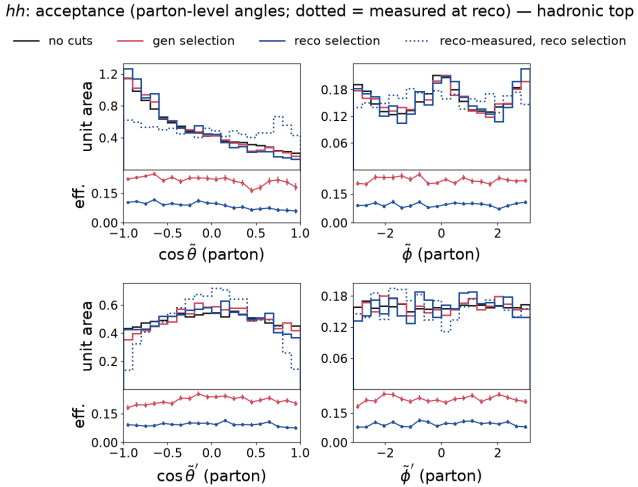
Acceptance, $t\bar{t}$: efficiency vs angle, cut by cut (hadronic top)



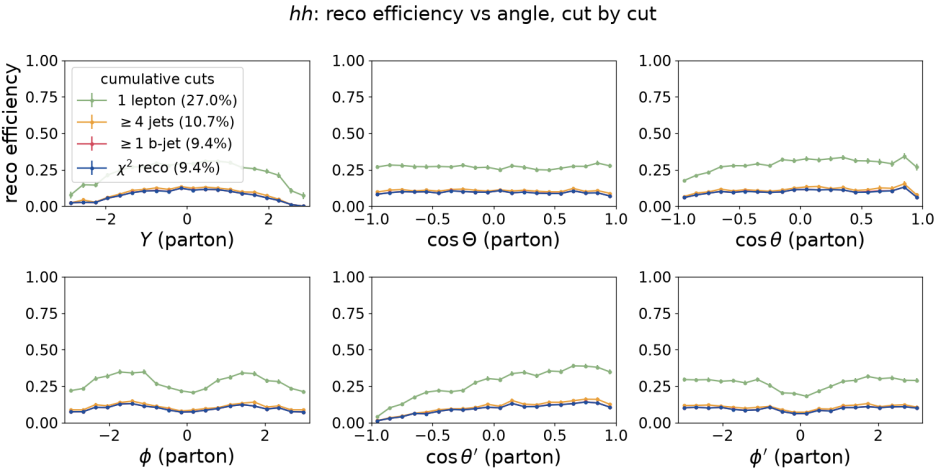
Acceptance, hh: no cuts vs after cuts (production, leptonic top)



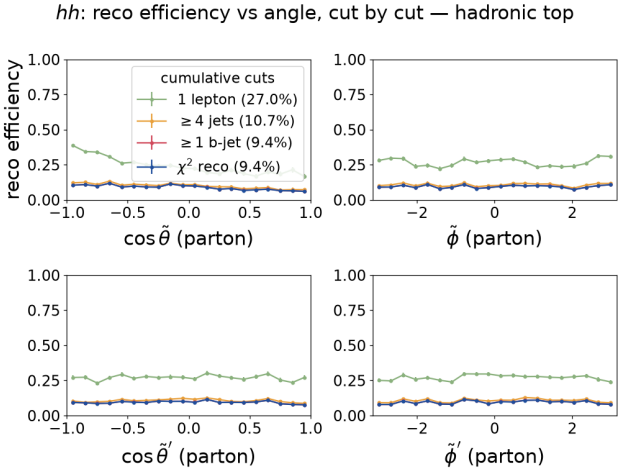
Acceptance, hh: no cuts vs after cuts (hadronic top)



Acceptance, hh: efficiency vs angle, cut by cut (production, leptonic top)



Acceptance, hh: efficiency vs angle, cut by cut (hadronic top)



Outline

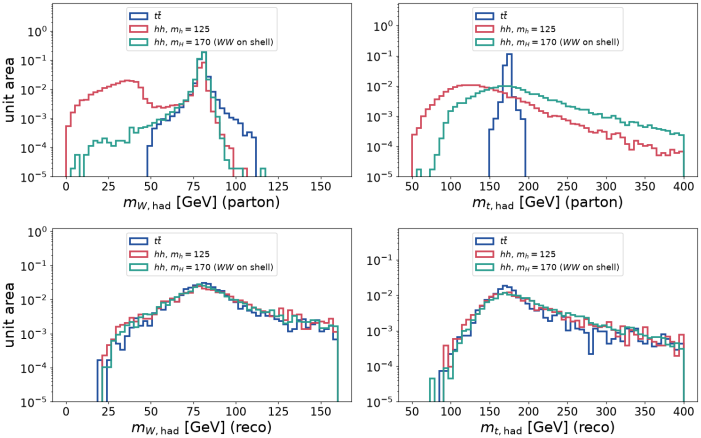
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2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Spherical-harmonic basis
9. Is the OSDE approach helping? (ML)
10. Why the separation washes out at gen level
11. Adding momentum to y: the resonance masses
12. Discriminator t (paper style)
13. Acceptance
14. Study A′: hh with on-shell W bosons ($m_H = 170$ GeV)
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16. Pileup
17. Level-1 trigger: a lookup-table version of angles + BW
18. Summary

Idea: remove the off-shell W handle and see what the angles do

- Same $hh \to b\bar{b}\,\ell\nu\,q\bar{q}'$ chain (`heft`, shapes only) with $m_H = 170$ GeV, just above $2m_W = 160.8$ GeV, so both Ws are on shell ($\Gamma_H = 0.38$ GeV); 20k events, same analysis chain, nothing retrained
- Expectation: the W Breit–Wigner factors go blind; the top factors still see m_{Wb} (the b comes from the other Higgs); after the χ^2 a genuinely on-shell hadronic W makes the signal more $t\bar{t}$-like; what is left is the angles: a scalar $\to WW$ decay is not $V-A$ (no zero at $\cos\theta' \to 1$), and the pseudo-top direction differs
- Selection efficiency (trigger / gen / reco): $hh(125)$ 51% / 22% / 9.4%; $hh(170)$ 73% / 44% / 20.0% (harder leptons from on-shell Ws)

$m_{W,\mathrm{had}}$ and $m_{t,\mathrm{had}}$: parton (top) and reco (bottom)

hh with on-shell *W*s: the *W* mass handle disappears, the top-mass handle stays



Which single variables differ at parton level (1-D likelihood-ratio AUC)

variable	$hh(125)$	$hh(170)$
$\cos\Theta$	0.610	0.613
$\cos\theta$	0.639	0.646
$\cos\theta'$	0.577	0.590
$\cos\tilde{\theta}$	0.634	0.642
$\cos\tilde{\theta}'$	0.577	0.590
$m_{W,h}$	0.799	0.545
$m_{t,\ell}$	0.968	0.924
$m_{t,h}$	0.972	0.904
$m_{W,\ell}$	0.797	0.550

AUC / signal efficiency at 1% $t\bar{t}$ efficiency, hh(125) and hh(170) vs $t\bar{t}$

score	parton 125	parton 170	gen 125	gen 170	reco 125	reco 170
angles, all terms	0.653 / 4%	0.638 / 4%	0.525 / 2%	0.547 / 2%	0.507 / 1%	0.534 / 1%
angles, $> 10\sigma$	0.670 / 8%	0.662 / 7%	0.536 / 2%	0.565 / 2%	0.518 / 2%	0.547 / 1%
BW, W masses only	0.992 / 67%	0.499 / 2%	0.628 / 2%	0.584 / 2%	0.574 / 1%	0.542 / 1%
BW, top masses only	0.999 / 98%	0.995 / 93%	0.684 / 3%	0.691 / 3%	0.592 / 2%	0.610 / 2%
BW, all masses	1.000 / 100%	0.985 / 78%	0.690 / 3%	0.680 / 2%	0.600 / 2%	0.600 / 1%
angles + BW	1.000 / 100%	0.986 / 80%	0.692 / 2%	0.685 / 2%	0.602 / 2%	0.606 / 1%

Same $f_{t\bar{t}}$ fits and mass factors as before. Parton: the four masses (W and top) with the PDG Breit–Wigner; gen/reco: $m_{W,h}$, $m_{t,\ell}$, $m_{t,h}$ with the two-number Cauchy.

Study A′: what it shows

- Parton: the W factors alone drop from 0.99 to 0.50, the top factors stay at 0.99: the mass handle was mostly the pseudo-top masses, not the W^*
- Angles alone at parton: 0.67 $\rightarrow$ 0.66; at reco: 0.52 $\rightarrow$ 0.55
- Angles + masses: gen 0.69 $\rightarrow$ 0.69, reco 0.60 $\rightarrow$ 0.61: unchanged. After reconstruction the W^* was never the handle (the χ^2 had already pulled m_{jj} to m_W); the pseudo-top masses and, less, the angles are
- So the method is not relying on the off-shell W: an hh-like signal with on-shell Ws is separated just as well, and selected twice as often

Study B: $t\bar{t}$ vs $H \to t\bar{t}$ (new physics in the production)

- An on-shell resonance: same masses as $t\bar{t}$, so the Breit–Wigner factors see nothing; the angles (production angle, spin correlations) and $m_{t\bar{t}}$ carry the difference
- Two masses: $m_H = 500$ GeV ($\Gamma = 25$ GeV) and $m_H = 350$ GeV ($\Gamma = 10$ GeV), just above the $2m_t = 346$ GeV threshold where only the spin correlations remain
- Two uses of $f_{t\bar{t}}$: as an anomaly score (low $t\bar{t}$ density), and its coefficients as a bump hunt in angle space ($\hat{c}_{\text{data}} - \hat{c}_{t\bar{t}}$)

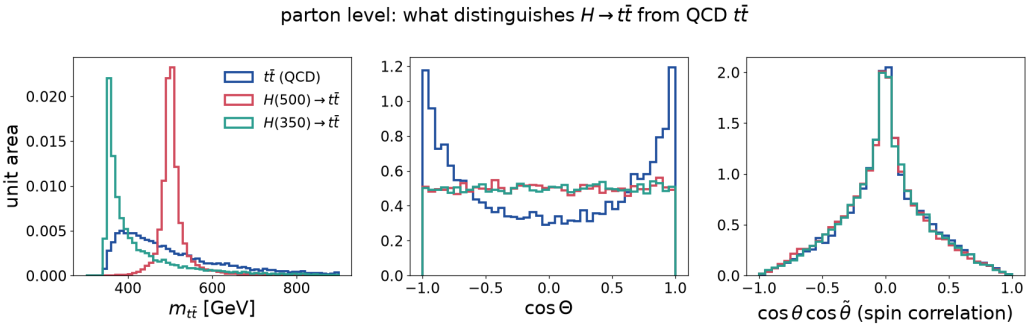
Outline

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2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Spherical-harmonic basis
9. Is the OSDE approach helping? (ML)
10. Why the separation washes out at gen level
11. Adding momentum to y: the resonance masses
12. Discriminator t (paper style)
13. Acceptance
14. Study A′: hh with on-shell W bosons ($m_H = 170$ GeV)
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16. Pileup
17. Level-1 trigger: a lookup-table version of angles + BW
18. Summary

Samples and what should differ

- $gg \to H \to t\bar{t} \to b\bar{b}\,\ell\nu\,q\bar{q}'$ (`heft` ggH, SM-like decays, signal only, no interference); MG5's automatic width is unusable here (4×10^5 GeV), so Γ_H is set by hand
- 20k events each; cross sections are LO heft normalizations and not used (shapes only)
- Scalar ($J = 0$): $\cos\Theta$ isotropic vs the t-channel forward peak of $gg \to t\bar{t}$; like-helicity tops at every mass, whereas QCD gg is like-helicity only near threshold; $m_{t\bar{t}}$ peaks at m_H on the bulk of the QCD spectrum (median 472 GeV)
- Near $2m_t$ the QCD $\cos\Theta$ flattens (S-wave) and the spin correlations become the main angular handle; our $H(350)$ sample has median $m_{t\bar{t}} = 384$ GeV (width and phase space), where QCD is still forward-peaked. The CMS threshold excess is this kind of object: a pseudoscalar ${}^1S_0$ state looks like QCD gg there, a scalar ${}^3P_0$ differs in the transverse correlations
- Test events: parton 10000 $t\bar{t}$, gen 4134 $t\bar{t}$, reco 2440 $t\bar{t}$; signals: $H(500)$ 20000, $H(350)$ 20000 (parton)

Parton level: $m_{t\bar{t}}$, $\cos\Theta$ and a spin-correlation proxy



Which variables differ (parton; 1-D likelihood-ratio AUC vs $t\bar{t}$)

variable	$H(500)$	$H(350)$
Y	0.515	0.499
$\cos\Theta$	0.615	0.610
$\cos\theta$	0.512	0.497
ϕ	0.501	0.498
$\cos\theta'$	0.498	0.504
ϕ'	0.506	0.499
$\cos\tilde{\theta}$	0.501	0.500
$\tilde{\phi}$	0.506	0.501
$\cos\tilde{\theta}'$	0.493	0.497
$\tilde{\phi}'$	0.503	0.502
$m_{t\bar{t}}$	0.880	0.734

Likelihood ratio from a 40-bin histogram fitted on half the events, tested on the other half, so non-monotonic differences count. $\cos\Theta$ and $m_{t\bar{t}}$ carry both signals; the single decay angles show nothing, as they must: spin correlations change the joint distribution, not the 1-D marginals.

$f_{t\bar{t}}$ as an anomaly score: $H(500)$ vs $t\bar{t}$

score	parton AUC	ϵ_S at $\epsilon_{t\bar{t}}$ 1% / 10%	gen AUC	ϵ_S 1% / 10%	reco AUC	ϵ_S 1% / 10%
angles, all terms	0.561 ± 0.004	0.8% / 11%	0.559 ± 0.006	1.3% / 12%	0.560 ± 0.006	0.8% / 12%
angles, $> 10\sigma$	0.555 ± 0.004	0.7% / 10%	0.562 ± 0.006	1.4% / 12%	0.565 ± 0.007	0.8% / 12%
BW masses	0.508 ± 0.003	1.0% / 10%	0.541 ± 0.005	2.3% / 15%	0.533 ± 0.007	2.2% / 13%
$m_{t\bar{t}}$ density	0.557 ± 0.004	0.1% / 1%	0.492 ± 0.006	1.1% / 9%	0.483 ± 0.007	0.9% / 8%
angles + BW	0.514 ± 0.003	1.1% / 10%	0.552 ± 0.005	2.2% / 15%	0.547 ± 0.007	2.2% / 14%
angles + $m_{t\bar{t}}$	0.529 ± 0.004	0.1% / 1%	0.522 ± 0.006	1.4% / 10%	0.515 ± 0.007	1.1% / 9%
angles + BW + $m_{t\bar{t}}$	0.493 ± 0.003	0.9% / 9%	0.533 ± 0.006	2.3% / 15%	0.527 ± 0.007	2.4% / 13%
dedicated supervised BDT (reference)	0.894	–	0.693	–	0.669	–

ϵ_S: signal efficiency when the cut keeps 1% (10%) of $t\bar{t}$; 1% (10%) = no enrichment. Reference: a BDT trained on this signal (angles + masses + $m_{t\bar{t}}$), what a dedicated search could do.

$f_{t\bar{t}}$ as an anomaly score: $H(350)$ vs $t\bar{t}$

score	parton AUC	ϵ_S at $\epsilon_{t\bar{t}}$ 1% / 10%	gen AUC	ϵ_S 1% / 10%	reco AUC	ϵ_S 1% / 10%
angles, all terms	0.579 ± 0.004	1.0% / 13%	0.549 ± 0.005	1.6% / 12%	0.542 ± 0.008	0.7% / 11%
angles, $> 10\sigma$	0.572 ± 0.004	1.3% / 12%	0.550 ± 0.005	1.2% / 11%	0.553 ± 0.008	0.9% / 11%
BW masses	0.512 ± 0.004	1.1% / 11%	0.537 ± 0.005	1.2% / 13%	0.509 ± 0.007	1.2% / 11%
$m_{t\bar{t}}$ density	0.467 ± 0.003	0.4% / 4%	0.505 ± 0.005	0.8% / 11%	0.495 ± 0.006	0.9% / 7%
angles + BW	0.520 ± 0.004	1.2% / 11%	0.546 ± 0.005	1.1% / 13%	0.522 ± 0.007	1.1% / 11%
angles + $m_{t\bar{t}}$	0.511 ± 0.004	0.6% / 5%	0.526 ± 0.006	1.2% / 11%	0.522 ± 0.007	1.0% / 11%
angles + BW + $m_{t\bar{t}}$	0.511 ± 0.004	1.3% / 10%	0.538 ± 0.005	1.3% / 13%	0.513 ± 0.007	1.4% / 11%
dedicated supervised BDT (reference)	0.764	–	0.638	–	0.577	–

ϵ_S: signal efficiency when the cut keeps 1% (10%) of $t\bar{t}$; 1% (10%) = no enrichment. Reference: a BDT trained on this signal (angles + masses + $m_{t\bar{t}}$), what a dedicated search could do.

Reading the anomaly-score table

- The masses see nothing (on-shell), as expected; the angles alone give a weak but real separation (0.55–0.57 at every level, from $\cos\Theta$), and the mass factors *dilute* it (the summed surprisal is dominated by the dimensions with the largest spread)
- $m_{t\bar{t}}$ as a density does not help although it is by far the most different variable (1-D AUC 0.88 / 0.73): the resonance sits where the $t\bar{t}$ density is high, and at threshold the signal even looks *more* $t\bar{t}$-like than $t\bar{t}$ (AUC 0.47). "Low density" finds rarities, not excesses
- Tail enrichment at $\epsilon_{t\bar{t}} = 1\%$ is the number that matters for an anomaly trigger or selection; $H(500)$: parton 0.7%, gen 1.4%, reco 0.8%; $H(350)$: parton 1.3%, gen 1.2%, reco 0.9%
- A dedicated supervised search does far better, because it knows the signal. The question for Study B is therefore not the score but whether the *coefficients* can find the signal without knowing it

The OSDE coefficients as a bump hunt in angle space

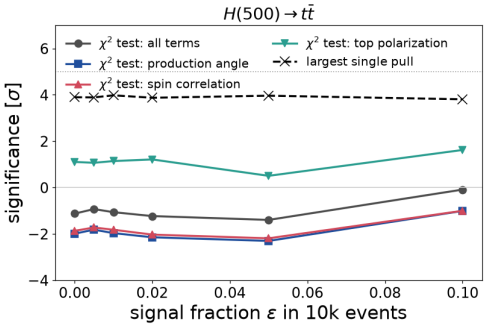
- Fit the 14 400 coefficients on the data (here: independent $t\bar{t}$, 10k events, plus a signal fraction ϵ) and compare with the $t\bar{t}$ reference (fit sample, 100k): pull $= (\hat{c}_{\text{data}} - \hat{c}_{t\bar{t}})/\sqrt{\sigma^2_{\text{data}} + \sigma^2_{t\bar{t}}}$ per term, over the 14 400 real degrees of freedom
- Model-agnostic: no signal hypothesis, no training; an *excess* shows as pulls on specific terms, which say what the physics is
- Calibration at $\epsilon = 0$: pull std 0.99, χ^2/ndf over all terms 14209/14400, $z = -1.1$: the two $t\bar{t}$ samples agree and the errors are right
- Tests: χ^2 over all terms and over each physics category (production angle, rapidity, W helicity, top polarization, spin correlation, SM-null terms), $z = (\chi^2 - \text{ndf})/\sqrt{2\,\text{ndf}}$; and the largest single pull (14 400 trials: 5σ global $\approx |\text{pull}| > 5.3$)

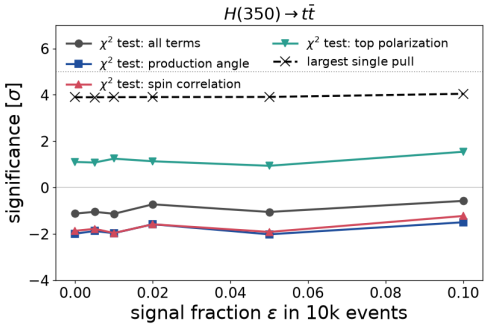
Coefficient test: result with 10k events, and how it scales

- Result with 10k events: no category χ^2 responds up to $\epsilon = 10\%$ (one term carries the signal, the rest of the block is noise); the single term that moves is $P_2(\cos\Theta)$, the isotropy of the scalar decay, with pull $H(500)$: 3.6, $H(350)$: 4.0 at $\epsilon = 10\%$, just at the noise ceiling (largest of 14 400 null pulls ≈ 3.9)
- The pull of a term grows as $\epsilon\sqrt{N}$: 5σ global ($|\text{pull}| = 5.3$) needs $\epsilon\sqrt{N} \approx H(500)$: 16, $H(350)$: 14, i.e. $\epsilon \approx 1.6\%$ at 10^6 events (0.16% at 10^8); the $m_{t\bar{t}}$ bump hunt is $\sim$7 times more sensitive to this signal ($S/\sqrt{B}$ in a $\pm\Gamma$ window)

Significance of the coefficient test vs signal fraction

OSDE coefficients as a bump hunt: $\hat{c}_{\text{data}} - \hat{c}_{t\bar{t}}$ (parton level)





$H(500)$ at $\epsilon = 10\%$: χ^2 per physics category

terms	χ^2 / ndf	z
all terms	14385 / 14400	-0.1
production angle	11367 / 11520	-1.0
rapidity	11488 / 11520	-0.2
W helicity	13974 / 14000	-0.2
top polarization	3508 / 3375	+1.6
spin correlation	10876 / 11025	-1.0
pol. x W helicity interference	13493 / 13500	-0.0

Which block of the expansion the signal lands in; fewer terms = less noise dilution = more sensitive.

Terms that respond, $H(500)$ at $\epsilon = 10\%$ (largest pull 3.8)

term	physics	|pull|	$\|\hat{c}\|/\sigma$ in $t\bar{t}$
$M^{11}_{0,-1} \times \tilde{M}^{12}_{1,1} \times P_2(\cos\Theta)$	$t\bar{t}$ spin correlation; top pol. x W helicity interference (lep); top pol. x W helicity interference (had); production angle	3.8	2
$M^{12}_{-1,1} \times \tilde{M}^{01}_{0,0} \times P_1(\cos\Theta) \times h_2(Y)$	top pol. x W helicity interference (lep); W helicity (had); production angle; rapidity	3.8	2
$M^{10}_{-1,0} \times \tilde{M}^{12}_{1,-1} \times P_1(\cos\Theta) \times h_4(Y)$	$t\bar{t}$ spin correlation; top polarization (lep, transverse); top pol. x W helicity interference (had); production angle; rapidity	3.7	1
$M^{01}_{0,0} \times \tilde{M}^{12}_{1,-1} \times P_2(\cos\Theta) \times h_4(Y)$	W helicity (lep); top pol. x W helicity interference (had); production angle; rapidity	3.7	2
$\tilde{M}^{12}_{1,-1} \times P_2(\cos\Theta)$	top pol. x W helicity interference (had); production angle	3.6	0
$M^{12}_{1,-1} \times \tilde{M}^{11}_{-1,0} \times P_2(\cos\Theta) \times h_4(Y)$	$t\bar{t}$ spin correlation; top pol. x W helicity interference (lep); top pol. x W helicity interference (had); production angle; rapidity	3.6	1
$P_2(\cos\Theta)$	production angle	3.6	115
$M^{11}_{-1,1} \times \tilde{M}^{02}_{0,0}$	top pol. x W helicity interference (lep); W helicity (had)	3.6	2

The last column is the term's significance in the $t\bar{t}$ reference: a large pull on a term that is zero in $t\bar{t}$ is new physics in a channel the SM does not have.

terms	χ^2 / ndf	z
all terms	14302 / 14400	-0.6
production angle	11292 / 11520	-1.5
rapidity	11450 / 11520	-0.5
W helicity	13871 / 14000	-0.8
top polarization	3502 / 3375	+1.5
spin correlation	10842 / 11025	-1.2
pol. x W helicity interference	13379 / 13500	-0.7

Which block of the expansion the signal lands in; fewer terms = less noise dilution = more sensitive.

Terms that respond, $H(350)$ at $\epsilon = 10\%$ (largest pull 4.0)

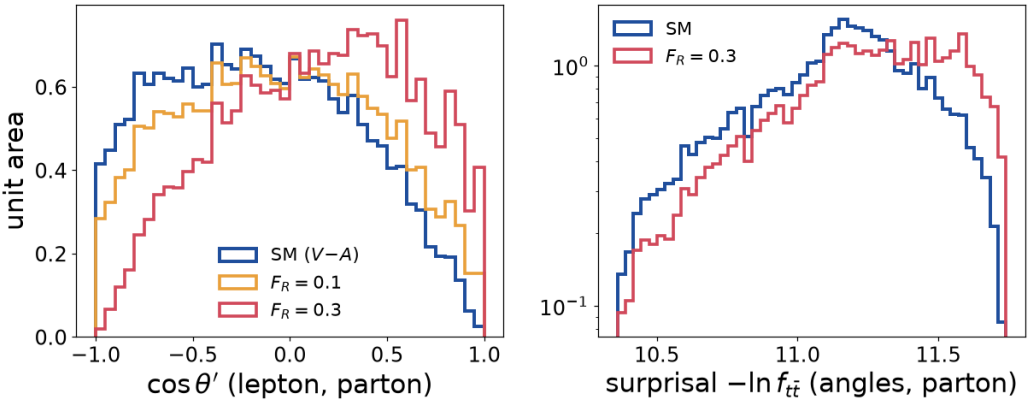
term	physics	|pull|	$\lvert\hat{c}\rvert/\sigma$ in $t\bar{t}$
$P_2(\cos\Theta)$	production angle	4.0	115
$M^{11}_{-1,1} \times \bar{M}^{11}_{1,-1} \times h_4(Y)$	$t\bar{t}$ spin correlation; top pol. x W helicity interference (lep); top pol. x W helicity interference (had); rapidity	3.8	1
$M^{10}_{-1,0} \times \bar{M}^{12}_{1,-1} \times P_1(\cos\Theta) \times h_4(Y)$	$t\bar{t}$ spin correlation; top polarization (lep, transverse); top pol. x W helicity interference (had); production angle; rapidity	3.8	1
$M^{12}_{-1,1} \times \bar{M}^{01}_{0,0} \times P_1(\cos\Theta) \times h_2(Y)$	top pol. x W helicity interference (lep); W helicity (had); production angle; rapidity	3.7	2
$M^{01}_{0,0} \times \bar{M}^{11}_{-1,1} \times P_2(\cos\Theta) \times h_3(Y)$	W helicity (lep); top pol. x W helicity interference (had); production angle; rapidity	3.7	2
$M^{10}_{-1,0} \times \bar{M}^{12}_{-1,0} \times P_3(\cos\Theta)$	$t\bar{t}$ spin correlation; top polarization (lep, transverse); top pol. x W helicity interference (had); production angle	3.7	1
$M^{01}_{0,0} \times \bar{M}^{12}_{-1,-1} \times P_2(\cos\Theta) \times h_4(Y)$	W helicity (lep); top pol. x W helicity interference (had); production angle; rapidity	3.6	2
$\bar{M}^{12}_{-1,-1} \times P_2(\cos\Theta)$	top pol. x W helicity interference (had); production angle	3.6	0

The last column is the term's significance in the $t\bar{t}$ reference: a large pull on a term that is zero in $t\bar{t}$ is new physics in a channel the SM does not have.

B2: a right-handed W in $t\bar{t}$ (new physics in the decay vertex)

- The textbook case for a density: the SM $(V-A)$ decay density vanishes at $\cos\theta' \to +1$, and a right-handed W helicity fraction $F_R > 0$ fills exactly that region. Constrained to $F_R \lesssim 0.03$ by the W helicity and single-top measurements, so a benchmark rather than a discovery channel
- Emulated by reweighting the independent $t\bar{t}$ sample with the parton-level W-decay distributions on both sides: $\rho(c) = \frac{3}{4}(1-c^2)F_0 + \frac{3}{8}(1-c)^2 F_L + \frac{3}{8}(1+c)^2 F_R$, F_0 kept, $F_L \to F_L - F_R$, $F^{\mathrm{SM}} = (0.703, 0.296, 0.001)$; gen and reco scores of the same events carry the parton weights
- Coefficient fit with event weights, $\hat{c} = \sum wB^* / \sum w$, with the weighted-variance error (effective number of events $(\sum w)^2 / \sum w^2$)
- Two uses of $f_{t\bar{t}}$ as before: the anomaly score (AUC, tail enrichment) and the coefficient test (pulls per term, χ^2 per physics category)

$\cos\theta'$ and the surprisal tail vs F_R (parton)



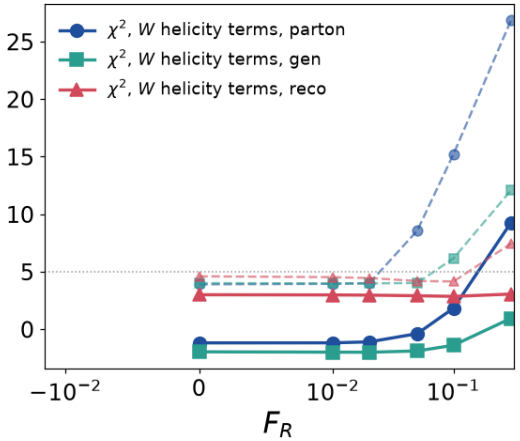
A right-handed W in $t\bar{t}$: where the density is in its element. Right: the surprisal of the score "angles ($> 10\sigma$)" of the next table.

Anomaly score: AUC / signal efficiency at 1% $t\bar{t}$ efficiency

score		parton	gen	reco
angles ($> 10\sigma$)	$F_R = 0.02$	0.507 / 1.2%	0.501 / 1.0%	0.501 / 1.1%
	$F_R = 0.05$	0.517 / 1.6%	0.504 / 1.1%	0.501 / 1.1%
	$F_R = 0.1$	0.534 / 2.2%	0.507 / 1.1%	0.503 / 1.3%
	$F_R = 0.3$	0.600 / 4.9%	0.522 / 1.3%	0.505 / 1.6%
angles + BW	$F_R = 0.02$	0.501 / 1.0%	0.500 / 1.0%	0.501 / 1.0%
	$F_R = 0.05$	0.502 / 1.0%	0.501 / 1.0%	0.503 / 1.0%
	$F_R = 0.1$	0.505 / 1.0%	0.502 / 1.0%	0.505 / 1.0%
	$F_R = 0.3$	0.513 / 1.0%	0.507 / 0.8%	0.513 / 0.8%
BW masses	$F_R = 0.02$	0.500 / 1.0%	0.500 / 1.0%	0.501 / 1.0%
	$F_R = 0.05$	0.500 / 1.0%	0.501 / 1.0%	0.502 / 1.0%
	$F_R = 0.1$	0.501 / 1.0%	0.501 / 1.0%	0.505 / 1.0%
	$F_R = 0.3$	0.502 / 0.9%	0.505 / 0.9%	0.511 / 0.8%

1% = no enrichment. The weighted-AUC errors are about ± 0.005 (parton) to ± 0.01 (reco).

The coefficient test vs F_R



Significance $[\sigma]$ of the coefficient test, per level (legend):

- solid: the χ^2 over the "W helicity" category, as z
- dashed: the largest single pull (same colour)
- dotted: 5σ

F_R axis linear below 0.01, logarithmic above. The single pulls are in the next table.

Coefficient test: pull of the $\cos\theta'$ term (M_{00}^{01}) vs F_R

	parton ℓ	had	max	gen ℓ	had	max	reco ℓ	had	max
$F_R = 0$	<4	<4	3.9	<4	<4	4.0	<4	<4	4.6
$F_R = 0.01$	<4	<4	3.9	<4	<4	4.0	<4	<4	4.5
$F_R = 0.02$	3.9	<4	4.0	<4	<4	4.0	<4	<4	4.4
$F_R = 0.05$	8.6	6.9	8.6	<4	<4	4.0	<4	<4	4.2
$F_R = 0.1$	15.2	13.6	15.2	6.2	<4	6.2	<4	<4	4.1
$F_R = 0.3$	26.9	26.1	26.9	12.1	7.3	12.1	7.4	4.9	7.4

ℓ / had: the term on the leptonic / hadronic side; max: the largest pull of all terms. Data: the independent $t\bar{t}$ sample (10k parton, 4.1k gen, 2.4k reco events) reweighted and fitted with the weights; reference: the fit sample. <4: not among the leading terms (the largest of 14 400 null pulls is ≈ 4). The χ^2 over the whole "W helicity" category is diluted by its thousands of product terms and only moves at $F_R = 0.3$ ($z = +9$ at parton); the single term is the sensitive statistic.

B2: what it shows (1/2)

- As an anomaly score the tail is real but weak: at parton level $F_R = 0.3$ gives AUC 0.60 and 5% of the signal in the 1% $t\bar{t}$ tail (a factor 5 enrichment); at $F_R = 0.05$ only 1.6%; at reco level the smeared $\cos\theta'$ leaves nothing (0.51). The masses dilute it, as for H
- The coefficient test is a different story: the $\cos\theta'$ term of the leptonic W (M_{00}^{01}, i.e. $\langle\cos\theta'\rangle = (F_R - F_L)/2$) has pull 8.6 at $F_R = 0.05$ and 15.2 at $F_R = 0.1$ with 10k parton events, the hadronic side follows; at gen (6.2 at 0.1 with 4.1k events) and reco (7.4 at 0.3 with 2.4k events) the smearing costs sensitivity but the term is the same
- Pull $\propto F_R\sqrt{N}$: 5σ at the current limit $F_R = 0.02$ needs ≈ 23k parton-level events (more after reconstruction). This is the W helicity measurement in disguise, with the physics labelled by the term

B2: what it shows (2/2)

- Contrast with $H \to t\bar{t}$: new physics in the decay changes the *shape* of the most significant terms of f (large pulls on few terms); a resonance in the bulk changes the production angle by a few per cent of its coefficient
- So: the score finds signals in SM-forbidden regions when they are large; the coefficients find small deviations in the SM terms. Both are model-agnostic; the second is the sensitive one for allowed new physics

Study B: conclusions

- As an anomaly *score*, $f_{t\bar{t}}$ is weak for an on-shell resonance (AUC 0.5–0.6), scalar or vector: the signal sits where $t\bar{t}$ is dense. A vector would change which angular terms differ (opposite-helicity correlations; top polarization if the couplings are chiral), not the size of the modulation
- As a test in *coefficient space*, the same fit is calibrated and physical: for $H \to t\bar{t}$ one term ($P_2(\cos\Theta)$) moves and the $m_{t\bar{t}}$ bump hunt is more sensitive; for new physics in the decay vertex the W-helicity terms move strongly and the coefficients are the sensitive test
- Next: the test at gen/reco level with the extended basis (needs the orders the reconstruction requires); a pseudoscalar A (the CMS threshold case); a vector (Z') with chiral couplings; realistic ϵ from $\sigma \times \mathcal{B}$

Study C: towards the detector and the trigger

- Pileup: the reco-level method (angles + masses) at $\langle\mu\rangle = 0$, 50 and 200 with the Delphes ATLAS pileup card
- Level-1 trigger: the same score as a lookup-table computation on a Versal FPGA; resources, latency, precision

Outline

1. Introduction
2. $t\bar{t}$ samples and validation
3. The ten angles
4. hh samples
5. $f_{t\bar{t}}$ by OSDE
6. How many coefficients matter?
7. Higher orders for gen and reco
8. Spherical-harmonic basis
9. Is the OSDE approach helping? (ML)
10. Why the separation washes out at gen level
11. Adding momentum to y: the resonance masses
12. Discriminator t (paper style)
13. Acceptance
14. Study A′: hh with on-shell W bosons ($m_H = 170$ GeV)
15. Study B: $t\bar{t}$ vs $H \to t\bar{t}$
16. Pileup
17. Level-1 trigger: a lookup-table version of angles + BW
18. Summary

Setup

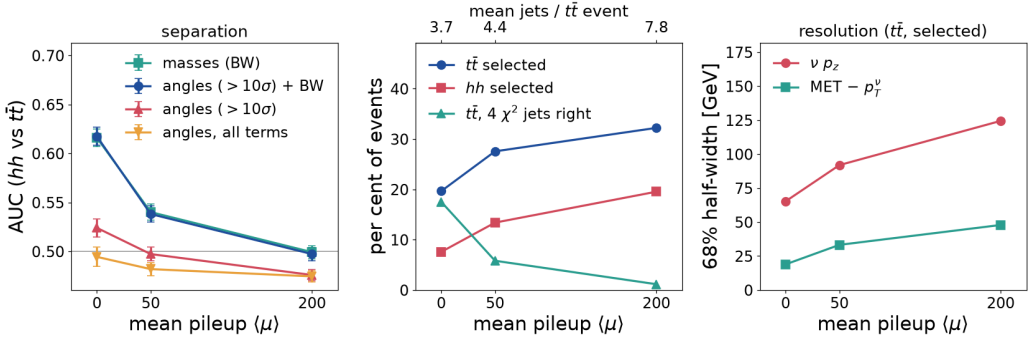
- Delphes ATLAS *pileup* card (pileup-subtracted jets, Rho, anti-k_t $R = 0.6$ as in the card, no pileup-jet rejection) at $\langle\mu\rangle = 0$ (control, same card), 50 and 200; min-bias library: 100k Pythia8 inelastic 13 TeV events
- Same hard events at every $\langle\mu\rangle$ (same seeds): $t\bar{t}$ fit 50k, $t\bar{t}$ test 10k, hh 20k generated; gen level is pileup-blind by construction, which gives an exact cross-check
- Identical analysis per $\langle\mu\rangle$ (`pileup/analyze_pileup.py`): reco-level $f_{t\bar{t}}$ refit, Cauchy mass factors from the fit sample, scores on the independent $t\bar{t}$ and on hh; the $\langle\mu\rangle = 0$ control differs from the no-pileup card (jet energy scale, calorimeter, MET), so compare within this set
- Cost: Delphes 17 / 75 / 560 ms per event at $\langle\mu\rangle = 0$ / 50 / 200 (single-threaded); produced in parallel jobs

Reco level versus pileup

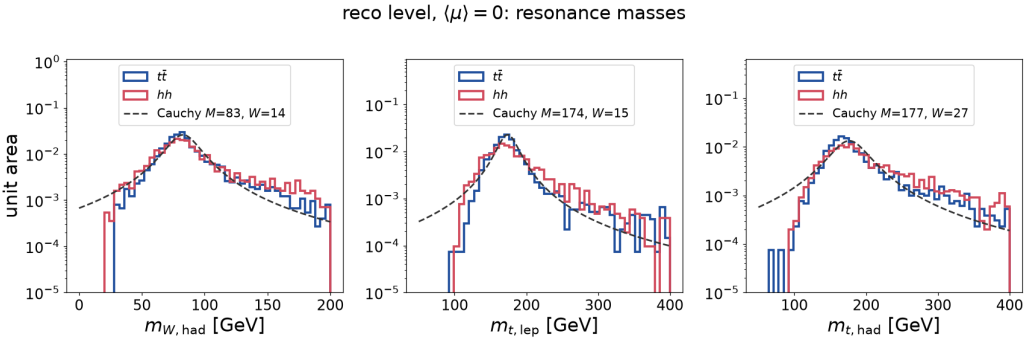
	$\langle\mu\rangle = 0$	$\langle\mu\rangle = 50$	$\langle\mu\rangle = 200$
$t\bar{t}$ fit / test events (reco)	10217 / 1965	13763 / 2755	15973 / 3221
hh events (reco)	1515	2677	3906
reco efficiency $t\bar{t}$ / hh	20 / 8%	28 / 13%	32 / 20%
jets per event ($p_T > 25$ GeV), $t\bar{t}$	3.73	4.43	7.75
b-tagged jets per event	1.15	1.09	1.01
all 4 χ^2 jets correct, $t\bar{t}$	17.5%	5.8%	1.1%
ν p_z / MET 68% half-width [GeV]	65 / 19	92 / 33	124 / 48
W: $m_{W,h}$ / $m_{t,\ell}$ / $m_{t,h}$ [GeV]	14 / 15 / 27	14 / 22 / 28	13 / 33 / 27
AUC angles, all terms	0.494	0.482	0.474
AUC angles, $> 10\sigma$	0.524	0.497	0.476
AUC masses (BW)	0.616	0.540	0.499
AUC angles ($> 10\sigma$) + BW	0.617	0.538	0.497
ϵ_{hh} at 1% / 10% $t\bar{t}$, angles + BW	1.4 / 17%	1.0 / 12%	1.1 / 10%
gen-level check (pileup-blind)	0.698	0.698	0.698

AUC, selection efficiencies and resolutions at reco level versus $\langle\mu\rangle$

Reco level versus pileup (Delphes ATLAS pileup card; same hard events at every $\langle\mu\rangle$)

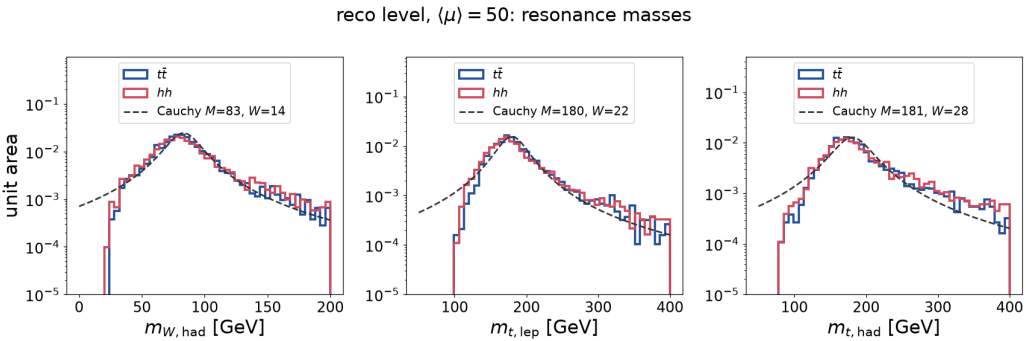


Resonance masses at reco level for $\langle\mu\rangle = 0$



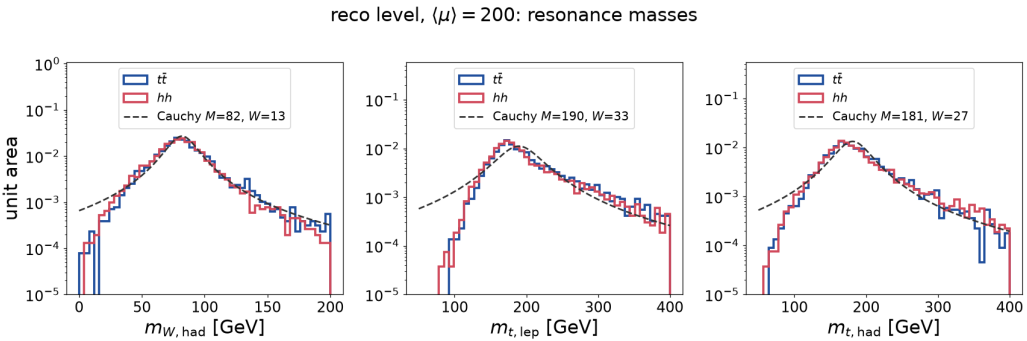
$t\bar{t}$ and hh, with the two-number Cauchy model.

Resonance masses at reco level for $\langle\mu\rangle = 50$



$t\bar{t}$ and hh, with the two-number Cauchy model.

Resonance masses at reco level for $\langle\mu\rangle = 200$



$t\bar{t}$ and hh, with the two-number Cauchy model.

Pileup: what it shows

- Selection: pileup jets pass the ≥ 4-jet cut, so the $t\bar{t}$ reco efficiency goes from 20% to 32% while the fraction of events with the right four jets goes from 18% to 1.1%: at $\langle\mu\rangle = 200$ the χ^2 almost never finds the $t\bar{t}$ system
- Resolutions: ν p_z 65 $\rightarrow$ 124 GeV, MET 19 $\rightarrow$ 48 GeV (and a MET bias of +36 GeV at $\langle\mu\rangle = 200$); only $W(m_{t,\ell})$ grows (15 $\rightarrow$ 33 GeV): with 7–8 jets the χ^2 always finds $t\bar{t}$-like masses, for $t\bar{t}$ and for hh alike
- The angle-only AUC even drops below 0.5 (0.476 at $\langle\mu\rangle = 200$): the reco f is dominated by wrong-jet combinations, so the genuine $t\bar{t}$ structure becomes the atypical part
- Separation: masses 0.616 $\rightarrow$ 0.499; angles + masses 0.617 $\rightarrow$ 0.497; the gen-level check is 0.698 / 0.698 (paired events, as it must be)
- Pileup-jet rejection (a card choice not made here) is the obvious lever: the physics loss is combinatorial, not resolution

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6. How many coefficients matter?
7. Higher orders for gen and reco
8. Spherical-harmonic basis
9. Is the OSDE approach helping? (ML)
10. Why the separation washes out at gen level
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12. Discriminator t (paper style)
13. Acceptance
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Can the score run in the ATLAS Phase-II Global Trigger (AMD Versal)?

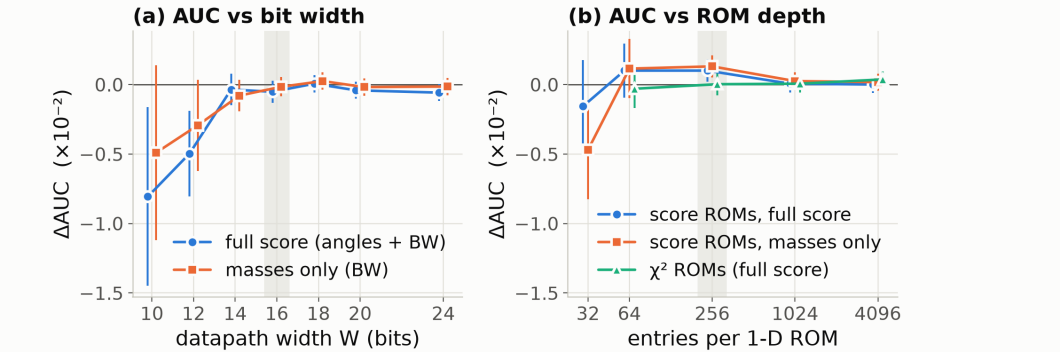
- Budget: 5% of the device and 10 μs. Inputs as L0 objects: 6 leading jets (p_T, η, ϕ, massless), one lepton, MET; quantized like the Run-3 TOBs; no b-tag
- The expensive part is the χ^2 jet assignment (180 assignments $\times$ 2 ν solutions), not the LUTs. Trick: for massless objects every hypothesis mass is a sum of shared pairwise invariants $m_{ij}^2 = 2p_i \cdot p_j$: 35 four-DSP dot products, 47 small χ^2 ROMs indexed by m^2 (no square roots), 360 adders and a comparator tree
- The ten angles need no Lorentz boosts: closed-form dimensionless expressions (aberration, ratios of the same invariants, perpendicular components), checked against `plotting/angles.py` to 10^{-10}
- Density and masses as 1-D ROMs (Legendre, Hermite, Wigner-d, $\cos$ / $\sin$, Breit–Wigner, $\ln$); the kept f has 31 terms
- Bit-true numpy emulation on the real reco samples; float chain reproduces the offline reference (angles + BW 0.602 vs 0.602)
- Resources and latency are analytic (no Vivado here); validation path: Vitis HLS / hls4ml (`study_l1/design.md` §8)

Recommended configurations on a Versal Premium VP1802

design	AUC	LUT	DSP	BRAM	latency (240 MHz)
masses only (χ^2 + BW); 6 jets, 16 bit, 256-entry ROMs	0.582	0.83%	1.45%	0.10%	0.43 μs
full score (+ angles)	0.583	1.21%	3.36%	0.26%	0.79 μs
offline reference (b-tag, truth down-type jet)	0.602	–	–	–	–

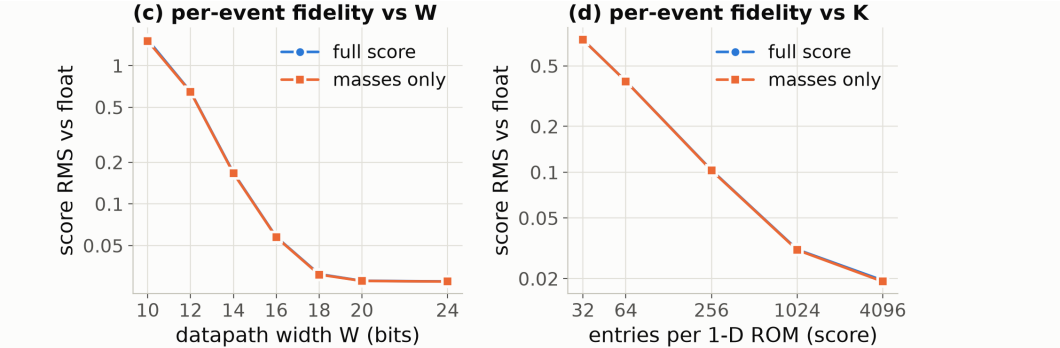
Fixed point costs nothing in AUC for datapath width ≥ 14 bits and ROM depth ≥ 64. The loss 0.602 $\rightarrow$ 0.583 is the missing b-tag at L1; restricting to 6 jets is free, 4 jets would cost another 0.018. Both designs are far inside 5% and 10 μs, and inside the 1.2 μs per-event budget of a GEP node. The angles add $+0.0004 \pm 0.0022$.

Precision: AUC vs datapath width W and ROM depth K (1/2)



N = 6 leading jets, no b-tag, nominal L1 inputs (pT 0.5 GeV, η 0.025, φ 2π/256, MET 1 GeV); (a) K = Kχ = 1024, (b) W = 18. (a, b) ΔAUC(hh vs $t\bar{t}$) relative to the float L1 chain (AUC 0.583 full, 0.582 masses only; offline reference 0.602 / 0.600); error bars: paired bootstrap ±1σ. Shaded: recommended W = 16, K = 256.

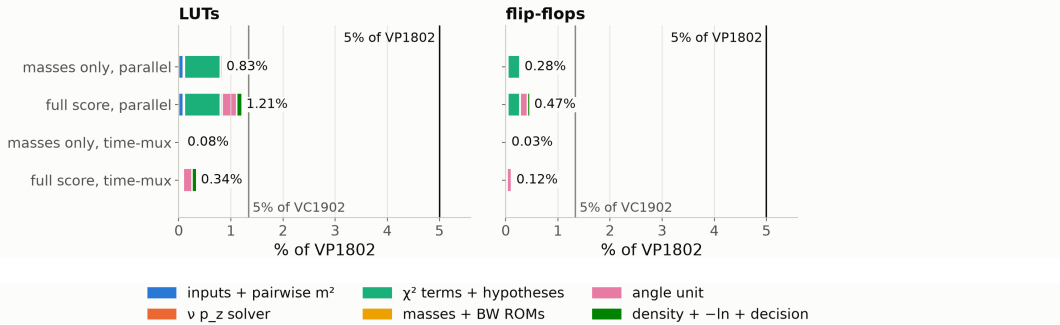
Precision: per-event fidelity vs W and K (2/2)



N = 6 leading jets, no b-tag, nominal L1 inputs (pT 0.5 GeV, η 0.025, φ 2π/256, MET 1 GeV); (a) K = Kχ = 1024, (b) W = 18.

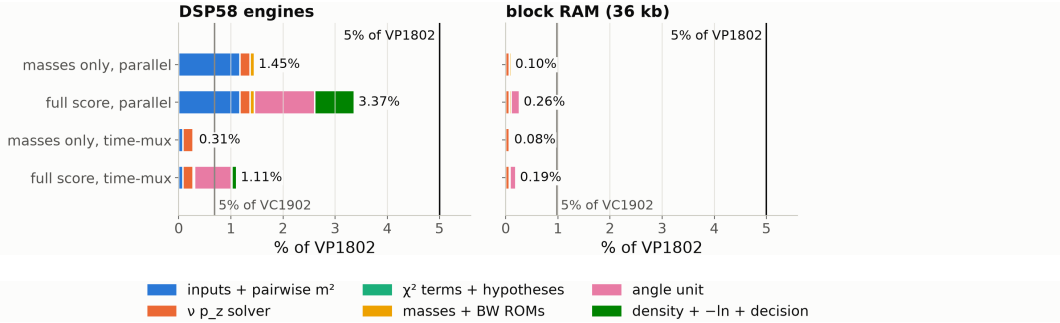
(c, d) RMS of the event score vs the same-input float chain, events with the same χ^2 pick (full and masses-only coincide). Same χ^2 pick vs W = 10, 12, 14, 16, 18, 24: 32%, 70%, 89%, 96%, 97%, 97%.

Resources by block, as a fraction of a VP1802 (1/2)



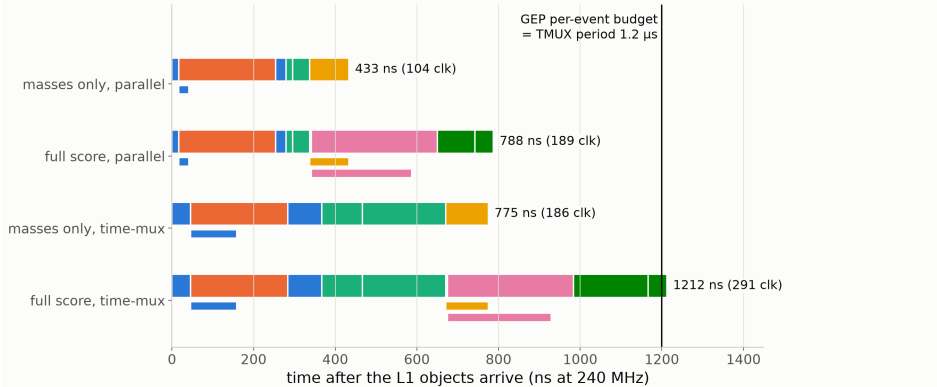
Analytic estimate (no synthesis): $N = 6$ jets, $W = 16$, $K = K_\chi = 256$; time-mux = loops serialized inside the 48-BC TMUX period (4 hypothesis lanes). LUT/FF include $\times 1.3$ for control and balancing. VP1802: 3.36 M LUT, 6.72 M FF, 14 352 DSP, 4 941 BRAM36; VC1902: 0.90 M, 1.80 M, 1 968, 967.

Resources by block, as a fraction of a VP1802 (2/2)



Analytic estimate (no synthesis): $N = 6$ jets, $W = 16$, $K = K_\chi = 256$; time-mux = loops serialized inside the 48-BC TMUX period (4 hypothesis lanes). LUT/FF include $\times 1.3$ for control and balancing. VP1802: 3.36 M LUT, 6.72 M FF, 14 352 DSP, 4 941 BRAM36; VC1902: 0.90 M, 1.80 M, 1 968, 967.

Pipeline schedule at 240 MHz



Thick bars: critical path; thin bars: blocks running concurrently. The whole ATLAS L0 budget of 10 µs is off scale (8–23× the totals).
Analytic pipeline depths (DSP58 3 clk, dot-product cascades, recip/rsqrt/sqrt units 12–18 clk, CORDIC W+1 clk); N = 6, W = 16, K = 256.
At 360 MHz all four designs scale by 2/3 (full score, time-mux: 808 ns).

L1: conclusions

- Feasible with large margins: $\sim$1% of the LUTs and 1.5–3.4% of the DSPs of a VP1802, 0.4–0.8 μs latency
- The hh-like tail survives quantization: at 5–10% $t\bar{t}$ acceptance, $\geq 95\%$ of the selected events are the same as in floating point; 91% of events get the same χ^2 assignment
- What limits the physics at L1 is the input, not the arithmetic: no b-tag, 6 jets, TOB precision. And the separation itself is weak at any precision (keeping 50% of hh keeps 38% of $t\bar{t}$)
- Recommendation: the masses-only design; add the angular term only if the parton-level physics of f is wanted in the trigger

Outline

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18. Summary

Summary (1/3): samples and angles

- $t\bar{t}$ samples (parton, gen, reco; ATLAS card, no pileup) produced and validated; hh (HEFT) likewise
- Ten angles as in the paper (diagram); parton level reproduces the SM W helicity fractions ($F_0 = 0.715 \pm 0.013$)
- Reco angles track parton angles when the jets are assigned correctly; combinatorics dominate the smearing
- ν p_z: 54 / 72 GeV (gen / reco, 68% half-width); the top-mass choice of the solution is not better than smaller $|p_z|$ (63% vs 65% closer to truth)

Summary (2/3): $f_{t\bar{t}}$ and hh

- $f_{t\bar{t}}$ from 14400 OSDE coefficients; AUC(hh vs independent $t\bar{t}$): parton 0.65, parton+trig 0.67, gen 0.53, reco 0.51
- Separation is lost at gen/reco because the $t\bar{t}$-hypothesis χ^2 makes hh look like $t\bar{t}$: e.g. a hadronic W^* (true median 32 GeV) is reconstructed near m_W; acceptance sculpts $\cos\theta'$, ϕ', Y
- Fit-sample vs independent-sample differences are entirely the self-inclusion bias (removed by leave-one-out)
- Only $\sim$50–350 coefficients matter: CV keeps 326, 385, 459, 342 (parton, gen, reco); simplified f has fewer negative values and the same AUC; parton level resolves $t\bar{t}$ spin correlations

Summary (3/3): $f_{t\bar{t}}$ and hh

- Paper basis has the right order at parton level, but gen/reco acceptance needs $j_{1,2}$ up to 13–16: the extended block basis (333, 498, 679, 323 terms after CV) describes every angle (χ^2/ndf $\approx$ 0.3–1.6; gen Y still imperfect); the hh AUC is unchanged
- Paper-style t: its shape depends on the choice of y_0 (AUC does not)
- Next: equiprobable-contour binning; truncated/extended f as default?; y_0 convention; 8-angle set; an f_{hh} for A–B testing; pileup